

자율주행이동체 실제

(9. Connected Vehicle)

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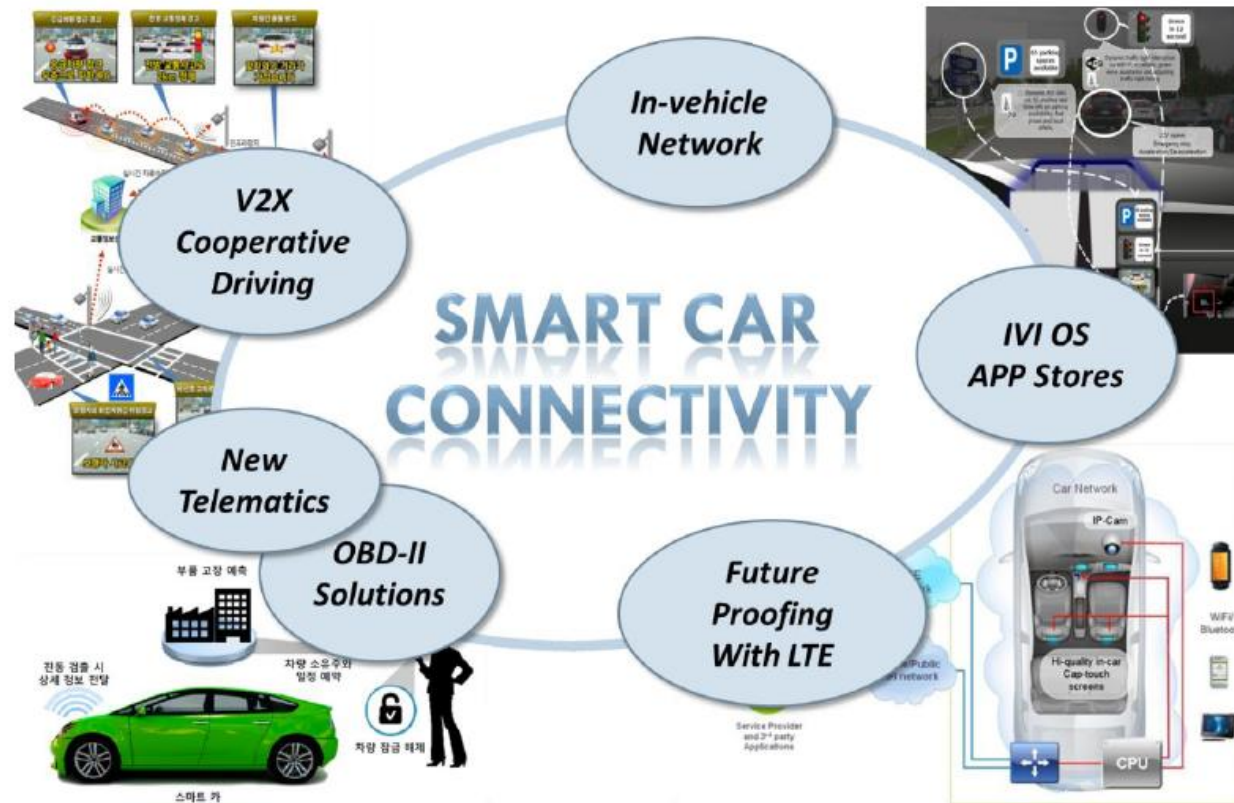
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Vehicle Connectivity

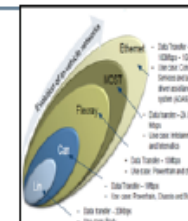
- 스마트카에서 안전기능과 편의기능을 극대화 하기 위한 차량간, 차량과 외부 인프라간, 차량 내부 제어기간의 정보 교환을 가능하게 하는 기술



In-Vehicle Network

Focus on Ethernet

- Ethernet/IP will co-exist with CAN, Flexray, MOST, etc.
- New E/E architecture base on Ethernet as a backbone
- Ethernet-enabled connected car with a Ethernet backbone



Cooperative Driving

Focus on Cooperative Driving

- Autonomous Automated Vehicle – Operates in isolation from other vehicles using internal sensors
- Connected Vehicle – V2X comm. Not automated (level 0)
- Connected Automated Vehicle – Leverages autonomous automated and connected vehicles
- KOREA – Automate vehicle & Highway Systems: AVHS



Future Proofing With LTE

Vehicle Cloud Services

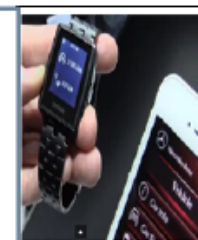
- Utilize the widespread growth of LTE for the connected car
- 3GPP – Vehicle Cloud Service base on 5G



Mobile Devices Connectivity

Focus on IVI OS And App Stores

- Competition in IVI OS and APIs space with more than 6 standards available (Google's official entry with the Open Automotive Alliance, Apple's iOS)



국제표준화 동향

- Networked Car 유무선 통신 플랫폼 기술 표준화
 - ISO TC22/SC31/WG2: 이더넷/IP 기반 차량 진단 프로토콜 표준화
 - ISO TC22/SC31/WG3: 차량 내부 네트워크 기술 표준화
 - ISO TC22/SC31/WG6: 차량 클라우드 서비스/원격 차량 진단 표준화
 - ISO TC22/SC32/WG8: 차량 기능 안전성 표준화
 - ISO TC204: V2X 통신 시스템 및 응용 서비스, 차량 제어/경고 시스템
 - IEEE 802.1 AVB Gen2: 동기식 이더넷 표준화

국내표준화 동향

- 자동차공학회 (KSAE)
 - 스마트카 안전/편의/네트워크 기술 표준화
- 한국지능형교통체계협회(ITS KOREA)
 - 차세대 ITS 시범사업을 위한 협력형 ITS 응용서비스 관련 단체표준개발
- 한국정보통신기술협회(TTA)
 - ITS/차량 ICT 표준화 그룹

자율주행차 표준화 포럼
(<https://avstandard.or.kr/>)

한국표준협회

국가기술표준원

- Demand for Connectivity
- Growth of In-vehicle Electronics
 - at least 40-50 ECUs on a Car, up to 100-140 ECUs on a premium car
- High Bandwidth Requirement
 - Camera-based applications, high-speed data comm.
- Need for Common Network Architecture
 - Increasing complexity of the vehicle network

- 관련 표준화 단체

- OPEN Alliance SIG
- IEEE 802.1 Ethernet AVB
- IEEE 802.3 RTPGE
- ISO TC22 SC31

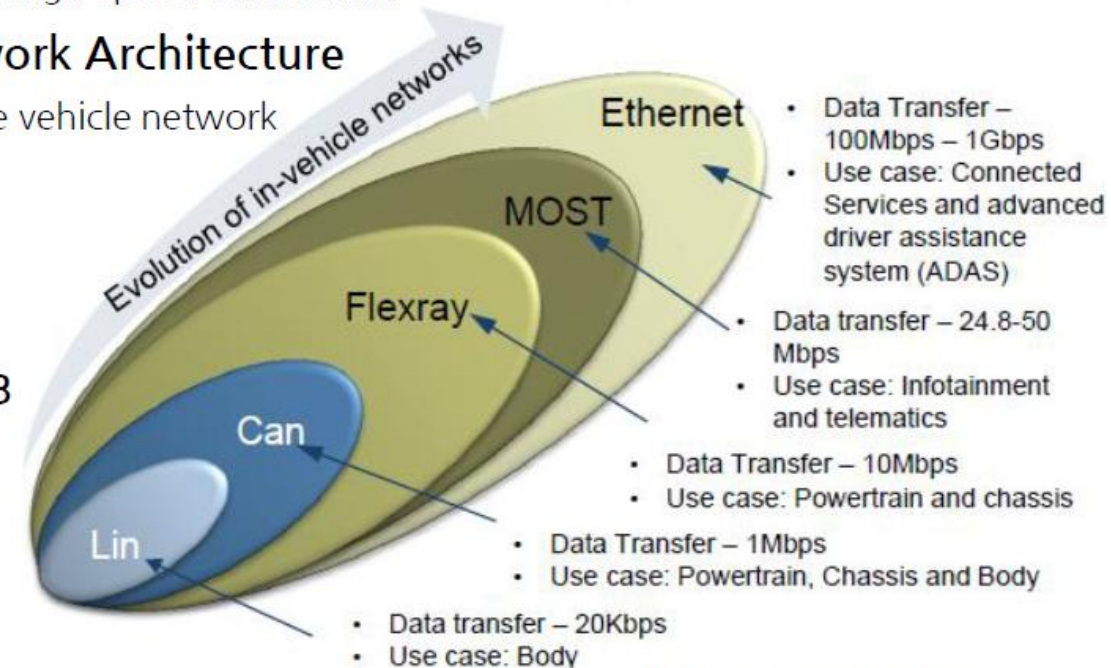
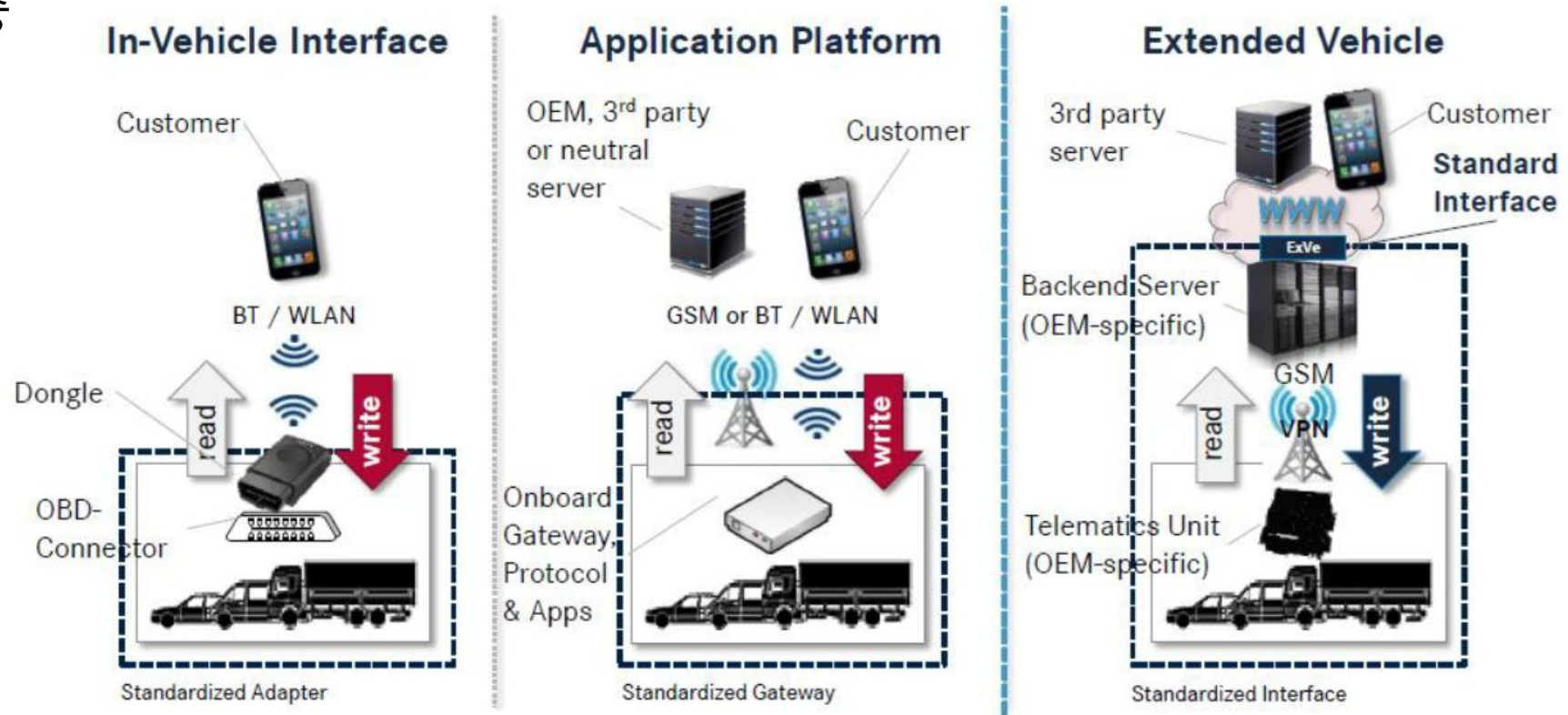


Image: Frost & Sullivan

Vehicle Interface

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- 스마트카는 단독형 시스템에서 협력형 시스템으로 발전
- 차량 내부 네트워크와 인프라 네트워크의 결합을 위한 IVN-V2X 게이트웨이 기술 개발 및 표준화 진행 중



ISO TC22 SC31 WG6 Extended vehicle

/ Connected & Automated Vehicle

- The path toward connected vehicles will ultimately lead to automated vehicles

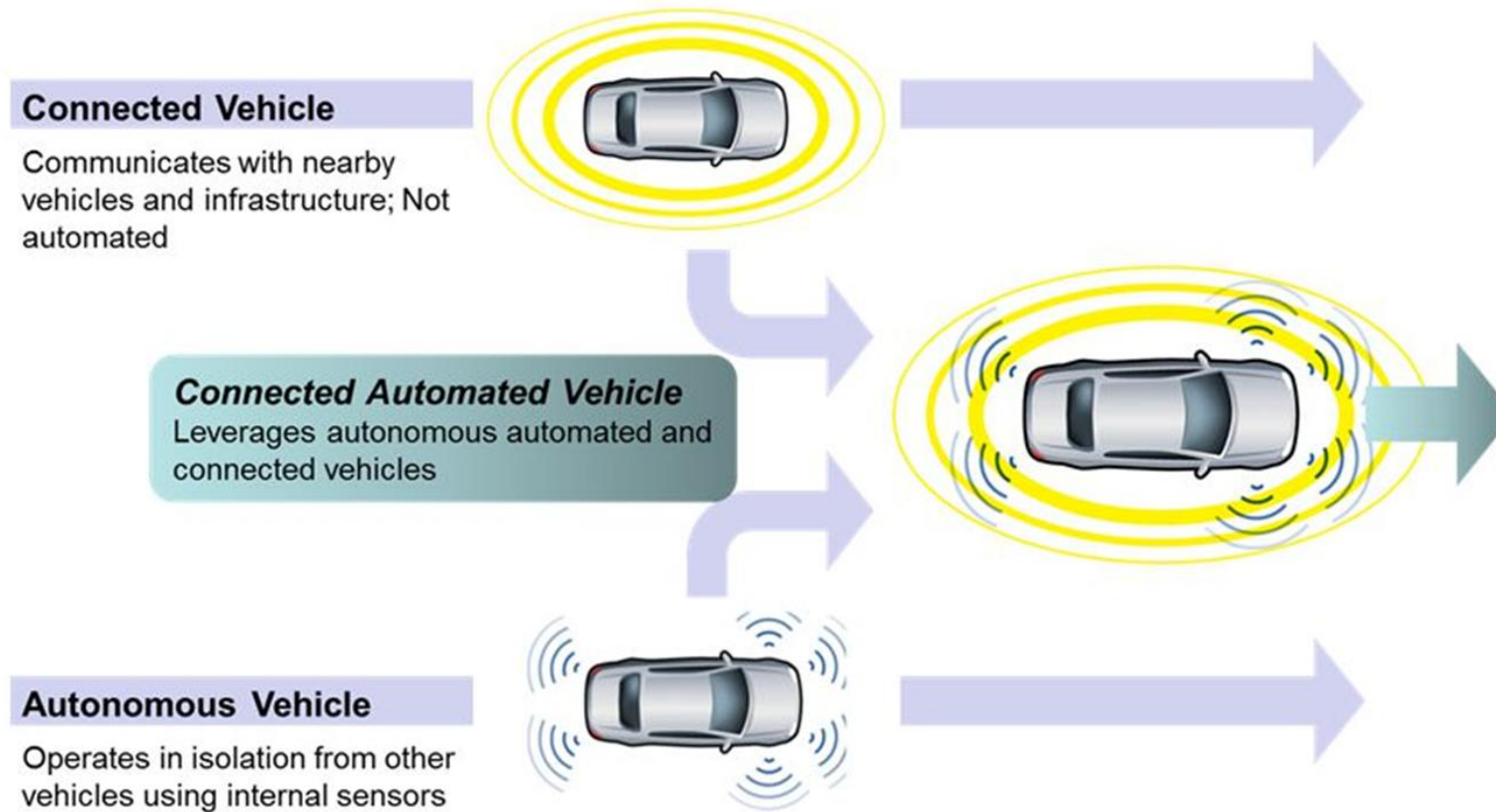
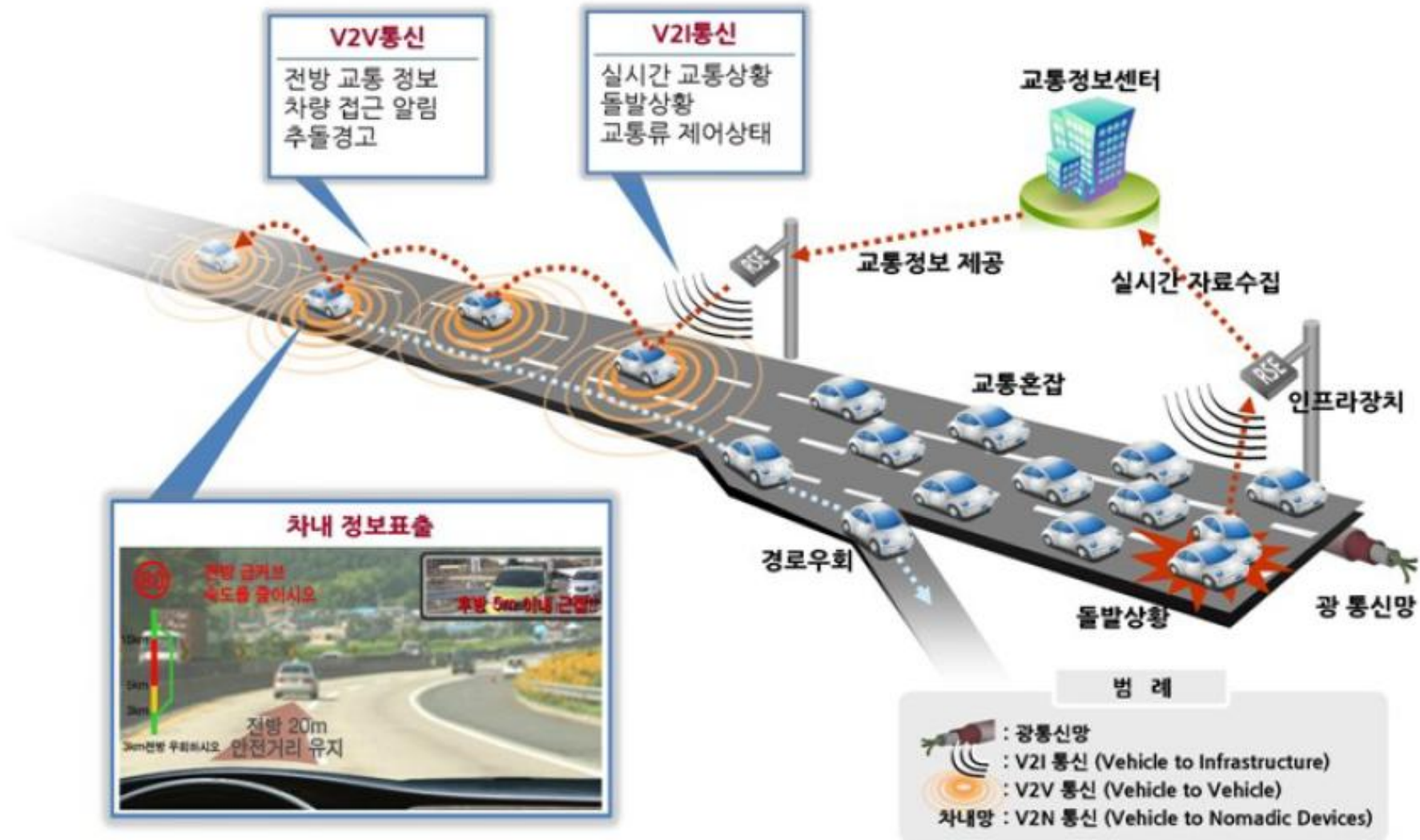


Image Source: Thinkstock/USDOT



V2X 개념도

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Cooperative Driving

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- Cooperative ITS Projects (ISO TC204 WG16, WG18)
 - 대전~세종 약 80km 구간 차세대 ITS 시범사업 진행 중
 - V2X 통신 기반으로 2차 사고 예방하기 위한 도로교통환경 구축
- Cooperative Safety 시스템 (ISO TC204 WG14)
 - V2V 기반 시스템: C-ACC, C-FCW 등
 - V2I 기반 시스템: Intersection Warning 등
- 주요 응용 서비스
 - Traffic Signal Violation Warning
 - Intersection Collision Warning
 - Roadwork warning
 - Road obstacle warning
 - Pedestrian collision risk warning
 - Forward collision warning
 - Emergency vehicle approaching warning

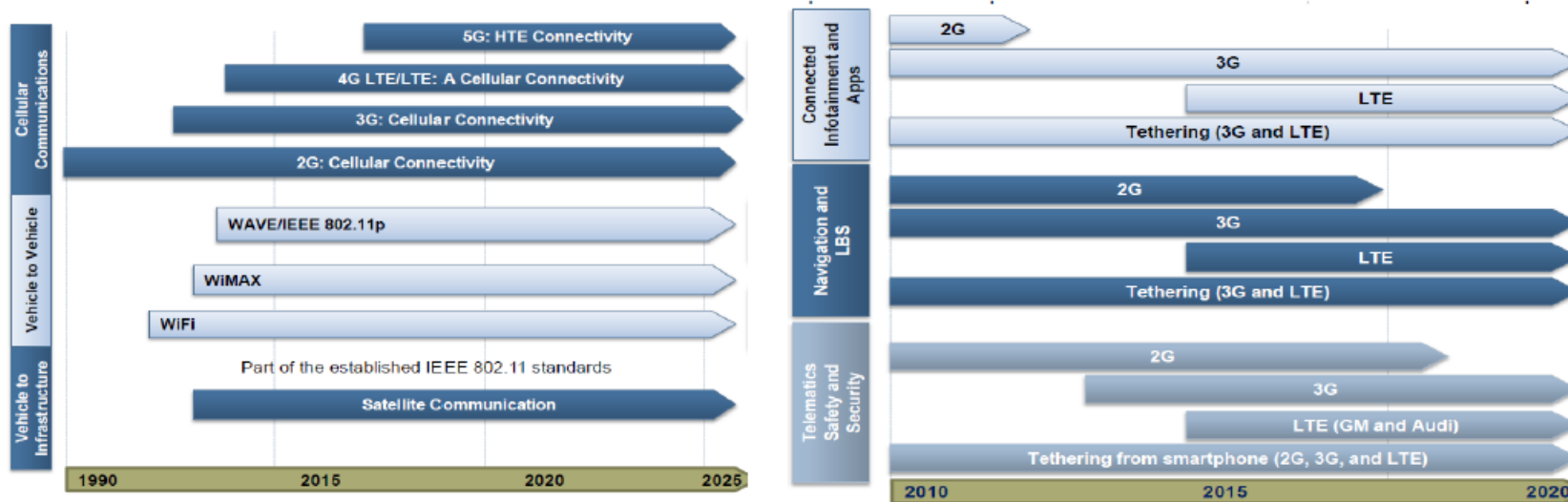
V2V APPLICATIONS			V2I APPLICATIONS		
Application name	Smart phone	AVN	Application name	Smart phone	AVN
Emergency Vehicle Approaching warning			Fallen Obstacle Warning		
Roadwork Warning			Animal or person Appearance Warning		
Collision Warning			Virtual VMS		
Emergency situation(SOS) Warning			ETC using WAVE		

Image- 2014년 ISO TC204 C-ITS Workshop in Vancouver

V2X Technology Roadmap

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- Cellular networks are becoming embedded into car (ex, eCall)
- Vehicle manufacturers are evaluating 802.11p for future short range, low latency V2X communication.
- Set to become the standard for short-range V2V
- Luxury automotive OEMs are leading LTE adoption
 - Audi – LTE option, BMW – Wi-Fi hotspot modules
 - LTE-based modem cost twice as much as 2G ones



Source: Frost & Sullivan

- 미국, 유럽, 일본의 C-ITS 및 자율협력주행 연구와 글로벌 자동차사의 자율주행 연구에서 WAVE와 셀룰러 통신 융합의 필요성을 제기하고 있으며 국제 표준 아키텍처 초안 완료
 - 미국은 2000년대 초부터 교통안전 서비스를 정의하고 안전서비스 제공을 위한 무선통신 중 WAVE 통신을 기반으로 단말기, 기지국을 개발하였고, NHTSA에서는 신규차량에 V2V 단말기 의무 장착 법제화 추진
 - 미국, 유럽은 상시 연결성 제공을 위해 WAVE와 이동통신을 융합한 Hybrid V2X 통신 시스템 개발

구 분	프로젝트 제목	비고
미 국	· CV Pilot Deployment(WAVE 1) – New York City, Tampa, Wyoming · CV Field Infrastructure Footprint Analysis (AASHTO, 2016.1)	WAVE, Other media WAVE, 4G, HD radio
유 럽	· Guide about for future C-ITS (ERTICO, 유럽ITS) · C-ITS Platform final report (EU 2016.01) · Autonet2030 · Compass4D, Companion Project · Ko-HAF research · scoop@F	WAVE, 2~5G, HD radio WAVE, Cellular WAVE, LTE WAVE, 3G/4G 3G/LTE WAVE, Cellular
일 본	· SIP – Automated Driving System · VICS	DSRC, Cellular FM radio, Infrared beacon

• 국내 표준은... **LTE-V2X**로 (2023.12.12)

- 4년을 끌어온 국가 차세대 지능형교통체계(C-ITS) 통신방식이 'LTE-V2X' 단일 방식으로 결정
- 실 도로 실증이 이뤄지지 않은 LTE-V2X 방식이 선정되면서 기준과 계획 마련 등은 지연될 전망
- 과학기술정보통신부와 국토교통부는 'C-ITS 단일 통신방식 결정 전문가 위원회' 결정에 따라 **LTE-V2X를 단일 통신방식으로 정하고, 기술기준 개정 등 후속조치를 추진한다고 발표(2023.12.12)**
- 오랜 기간 동안 실증을 거치고, 무료 이용이 가능한 WAVE 진영과 향후 5G-V2X를 고려했을 때 기술 확장성이 있는 LTE-V2X 진영의 논쟁으로 국내 C-ITS 서비스는 인프라 투자를 미룬 채 정체
- 이번 통신 방식 결정에 따라 일반 자동차 운전자들의 안전과 함께 자율주행자동차 안전성 확보에 필요한 C-ITS 서비스 투자가 본격화될 것으로 예상
- 과기정통부가 통신방식을 결정하면, 국토교통부가 그 방식에 따라 인프라 구축 사업을 추진하는 구조이며, LTE-V2X 방식에 대한 시험 결과는 있지만, 실 도로 실증은 미흡한 상태
- 국토부는 내년부터 실제 도로에서는 몇 km 간격으로 설치해야 할지, 터널에서는 어떤 방식으로 끊김을 보완 할지 등을 실증 테스트를 거쳐 기준을 마련할 방침. 전국 주요 도로에 기 설치된 WAVE 시설을 어떻게 활용할 지도 계획 필요

- 미국, 독일, 일본 등은 자율주행 및 Connected Car 를 중심으로 자동차와 정보통신, 에너지, 서비스 산업을 융합한 Smart Car 산업을 신성장 동력으로 육성
- Smart Car 기술은 첨단 센서 및 전자제어장치에 기반을 둔 차량 전자화 및 지능화 기술을 중심으로 발전하고 있으며, 최근에는 첨단 통신기술을 융합한 차량 간, 차량-인프라, 차량-클라우드 등 V2X 통신 기술을 통한 Connected Car 기술로 확대
- Connected Car 통신 기술은 자율주행 서비스의 고도화 및 다양화로 3G/4G 통신을 넘어 5G 통신으로 급속도로 발전
- Qualcomm은 Connected Car 의 첨단 통신기술의 융합을 위해 WAVE 통신을 포함하고 LTE 등의 이동통신 및 GNSS, Wi-Fi, BLE(Bluetooth Low Energy) 등의 통신 기술을 가변적으로 사용할 수 있는 Reference 플랫폼을 개발. 특히, 이 플랫폼은 기술발전에 따른 확장성이 고려되는 등 차후 차량통신 기술로 편입이 예상되는 5G 통신기술까지 지원 가능할 것으로 예상

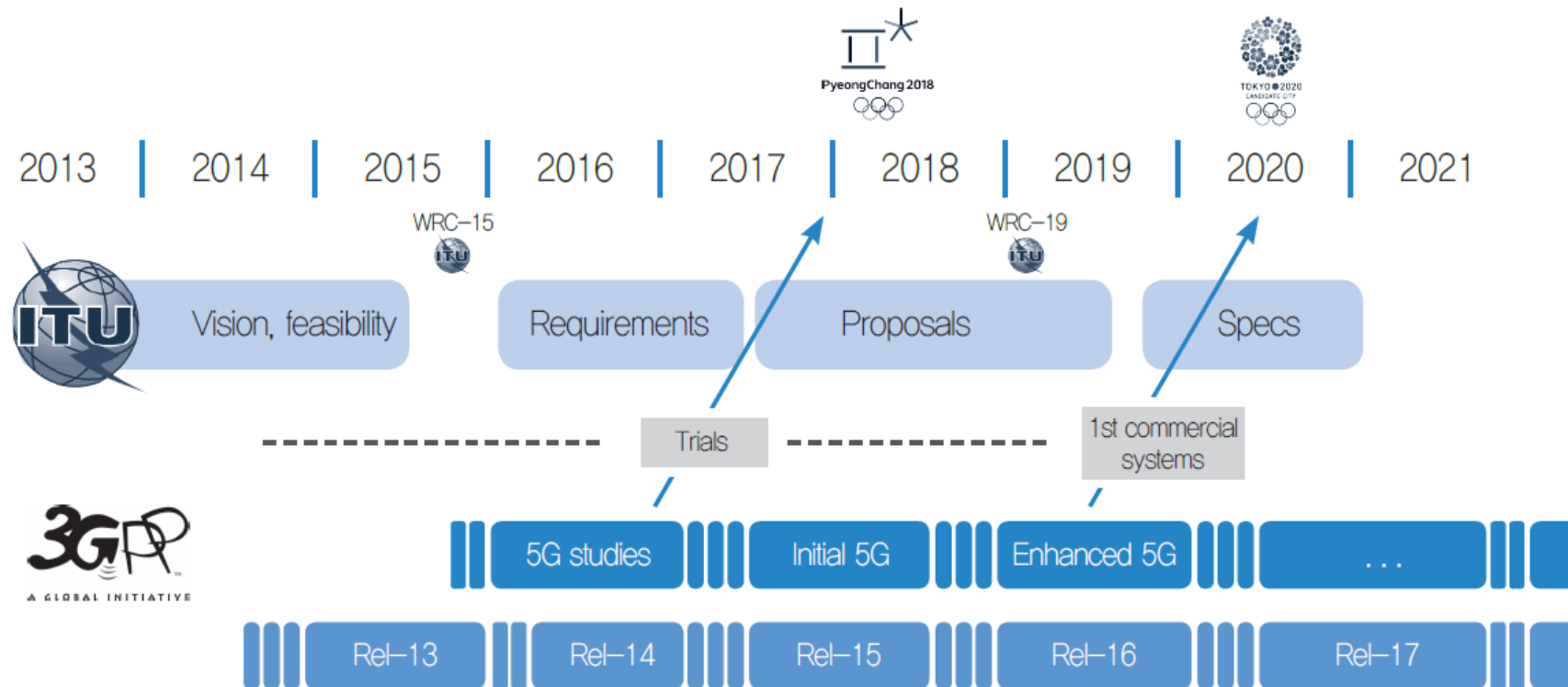
- LTE 기반 V2X 통신기술에 대한 표준은 3GPP Release14 버전에서 작업 중이며, 2018년 완료됨
- 대표적으로 LTE-V2X 기술은 화웨이, 노키아, 퀄컴 등 통신기술 기반 글로벌 기업 등에서 표준화 활동, ITS 세계대회 참여 등 기술 개발에 적극적

항목	WAVE (IEEE 802.11p)	LTE-V2X	5G-V2X (LTE-eV2X, NR-V2X)
Data Rate	최대 27Mbps	최대 100Mbps	최대 20Gbps (eMBB)
Reliability	95~99%	95~99%	99.9~99.999% (URLLC)
Latency	<100ms	<100ms	<10ms (URLLC)
Density	가능	가능	적합(제공km당 10^6대:mMTC)
Mobility	가능(최대 200km/h)	가능(160km/h)	가능(최대 500km/h)
Positioning	부적합(<50m)	부적합(<50m)	가능(<0.1m)
Coverage	가능 (평균 250~300m)	적합 (대략 수 킬로미터)	적합 (대략 수 킬로미터)
V2I & V2N	가능	적합	적합
차량 트래픽 혼잡 상황 대응	CSMA-CA 적용으로 전송 성능 열화	• 효율적 자원 할당으로 높은 트래픽 용량 제공 • 기지국 SPS 스케줄링 지원 시 혼잡 상황 해소 지원 가능	

5G-V2X 기술 동향 (KT)

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- 국내의 KT는 5G기술을 통해 커넥티드카 기반 자율주행차를 위한 초고속 무선 데이터 전송을 이룰 수 있도록 2018년 평창 올림픽에서 시범 서비스를 수행하고, 2020년 상용화를 시작할 것이라고 발표하였지만, 구체적인 5G-V2X 서비스 제공을 위한 개발 계획은 아직도 미정



5G Values



- Three Key Services of SKT



5G Connected Car Trial



- BMW-SKT Collaboration



5G Network

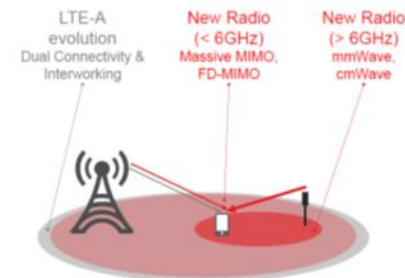
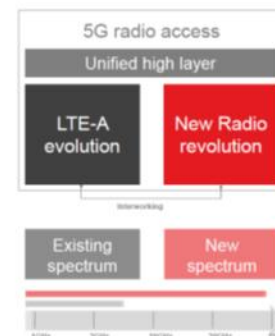


- 100x-1000x Speed
- 10x Lower Latency
- 10x-100x Connected Devices
- Virtualization
- Efficiency & Reliability

5G Radio



- New Radio + LTE A Evolution



/ 자율주행 V2X Requirements

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V2X category	Comm. type	Latency	Data rate	Reliability
Safety and traffic efficiency	V2V, V2P	~100 ms	Not a concern	In study
Autonomous driving	V2V, V2N, V2I	~1 ms	10 Mbps (downlink/downlink)	~100%
Tele-operated driving	V2N	~20 ms (E2E)	Uplink: 25 Mbps for video and sensors data Downlink: 1 Mbps for app.-related control and command	99.99%
Vehicular Internet and infotainment	V2N	~100 ms (web browsing)	0.5 Mbps (web browsing) Up to 15 Mbps for UHD video streaming	Not a concern
Remote diagnostics and management	V2I, V2N	N.A	Not a concern	Not a concern

	Basic safety services	Platooning	Advanced Driving	Remote Driving	Extended Sensor
					
Latency	≤ 20ms	≤ 10ms	≤ 3ms	≤ 20ms	≤ 3ms
Reliability	95%	99.99%	99.999%	99.999%	99.999%
Data rate	≤ 74 Mbps	65 Mbps	53 Mbps	25 Mbps	1 Gbps

15개 안전 서비스

*각 서비스 이미지를 클릭하시면 예시 동영상을 볼 수 있습니다.

1 위치기반 데이터 수집 V2I	2 위치기반 교통정보 제공 I2V	3 요금징수시스템 I2V	10 옐로우버스 운행 안내 V2V	11 스쿨존 속도제어 I2V	12 보행자 충돌방지 경고 I2V
4 도로위험구간 정보제공 I2V	5 노면 기상정보 제공 I2V	6 도로작업구간 주행지원 I2V	13 차량 충돌방지 지원 V2V	14 긴급차량 접근경고 V2V	15 차량 긴급상황 경고 I2V
7 교차로신호위반 위험경고 I2V	8 우회전 안전운행 지원 I2V	9 버스 운행관리 V2I			

미래차 산업발전 전략

비전

2030년 「미래차 경쟁력 1등 국가」로 도약

- 2030년 5-F(ree) 사회 구현 -

* 사고(Accident), 장벽(Barrier), 정체(Congestion), 지연(Delay), 배출(Emission)

목표

◆ 전기·수소차 보급* 세계 1위 국가, 세계시장 점유율 10% 달성

* 전가수소차 판매비중(%): ('19) 2.6 → ('20) 4.9 → ('22) 9.9 → ('25) 18.3 → ('30) 33.3

◆ 전국 주요도로의 완전자율주행(레벨4) 세계 최초 상용화('27)

* '21년 레벨3 자율차 출시 → '24년 레벨4 일부 상용화 → '27년 레벨4 전국 상용화

추진전략

◆ 친환경차 기술력과 국내보급 가속화를 기반으로 세계시장 적극 공략

* 이용자 편의제고 등 국내보급 확대 전략과 성능중심 세계시장 진출전략 병행

◆ 완전자율주행 법제도인프라(주요도로)를 세계에서 가장 먼저 완비('24)

* 레벨3→레벨4 단계적 추진전략에서 레벨3+4 동시 기반조성 전략으로 전환

◆ 민간투자(60조원)를 기반으로 개방형 미래차 생태계로 신속 전환

정책과제

친환경차 세계시장 선도 ① (생산·수출) 가격·성능 혁신 차종 다변화 등 글로벌경쟁력 확보
② (국내보급) 보조금 등 다양한 혜택 및 충전인프라 구축

자율주행차 미래시장 선점 ① (인프라) 자율주행 4대 국가 핵심인프라 구축('24)
② (제도) 자율차 제작·성능검증·보험·보안체계 마련('24)
③ (기술) 완전자율주행 상용화('24) 및 기술강국 도약('27)

미래차 서비스 시대 준비 ① (커넥티드) 서비스 걸림돌 제거하여 조기 확산
② (자율주행) 교통약자 지원 등 3대 선도서비스 추진

산업생태계 대전환 지원 ① (부품) 부품기업의 미래차 전환 가속화 지원
② (신사업) 개방형 협력을 통한 중소·중견기업 기회 확대
③ (소재·부품) 핵심 소재·부품 자립도 50 → 80%로 제고

기대효과

[기술력] 자율주행 기술선도국 도약, 세계최고 전기·수소차 기술력 확보

[안전] 교통사고 사망자 3,781명 → 1천명 이하(△74%), 교통정체 30% 저감

[환경] 온실가스 30% 감축, 미세먼지 11% 저감

Source: 범부처 미래자동차 산업발전 전략('19.10)

2 자율주행차 미래시장 선점

◆ '27년 전국 주요도로 완전자율주행(레벨4) 세계최초 상용화를 위하여 법·제도, 인프라(주요도로)를 세계에서 가장 먼저 완비('24)

- ① (인프라·제도) 완전자율주행 통신인프라 등을 전국 주요도로에 완비('24), R&D → 상용화의 발전단계별 규제정비·未備규정 마련('24)
- ② (자동차 개발) 완성차사는 차량개발·출시를 최대한 가속화하고, 정부는 차량 출시와 연동하여 부품국산화 등 산업생태계 지원

※ 기술발전에 따라 제도개선을 단계적으로 추진

구분	'19~'20	'21~'22	'23~'25	'26~'30
4대 인프라	- 통신망식 연구반('20) - 고속도로(5,500km) 정밀지도 구축('19) - 수도권 교통관제 시스템 10개소('20) - 신호등, 표지 등 인식도 개선작수('20)	- 통신망식 결정('21) - 주요거점 교통관제 시스템 34개소('22) - 인식도 강화 시범사업('22)	- 차량통신망 전국구축('24) - 신호등, 표지 등 전국 30% 통일('24)	- 전국 도로 11만km 정밀지도('30) - 교통관제시스템 전국 확대('30) - 교통안전시설 전국 구축('30)

○ (통신시설) “車↔車, 車↔도로” 간 무선통신망(V2X: Vehicle To Everything)을 전국 주요 구간에 구축하여 차량센서 기능 보완(~'24)



※ “車-도로”, “車-車” 간 통신으로 차량 센서(눈)의 인지 기능을 보완하여, 안전한 주행을 돕는 필수 인프라 (C-ITS: Cooperative-Intelligent Transportation System)
* 카메라 센서 인식거리는 수백 미터 vs 무선통신 활용시 수천 미터 앞 및 사각지대 신호전달 가능

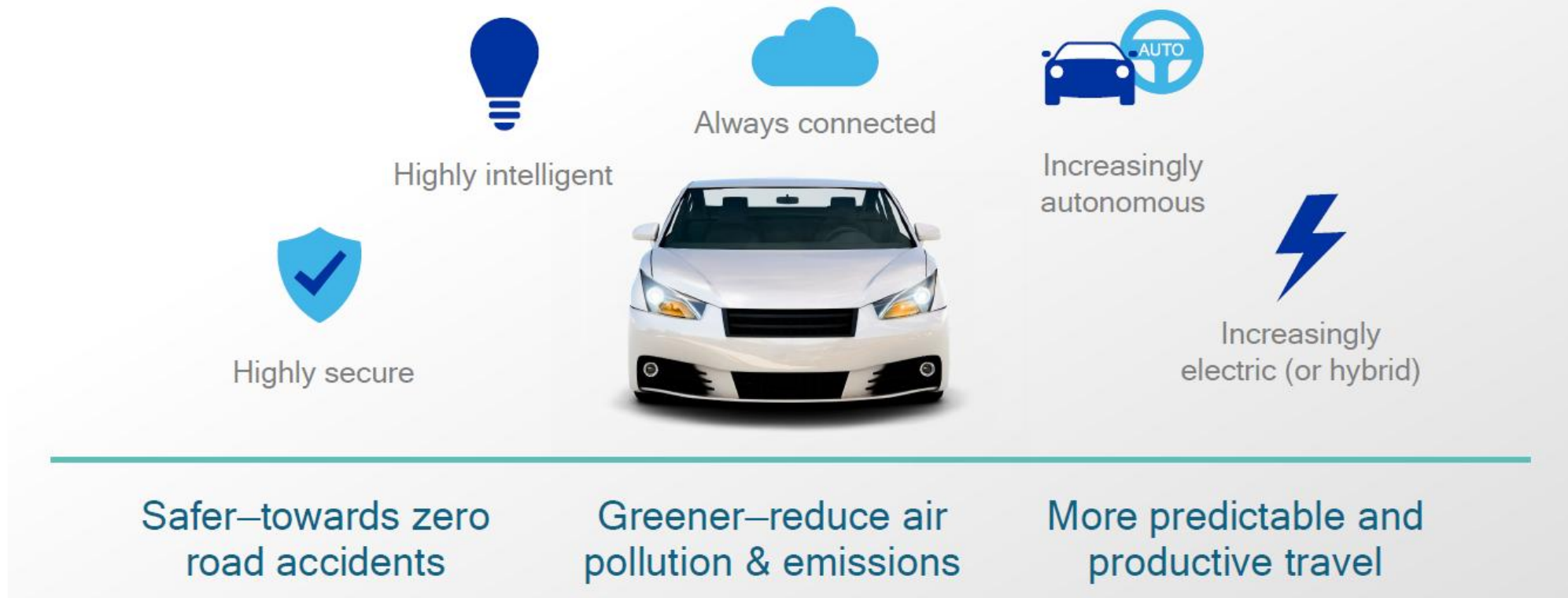
- 국제표준·주요국 정책동향 등을 고려하여 현재 미정인 국내 통신망식 결정 추진 ('21, 국토부, 과기정통부)

* WAVE / 5G-V2X / 병행 등 결정을 위해 부처·업계 등 공동연구반 운영(~'20)

- 도로시설·교통량 등을 고려하여, 통신시설 우선 구축 구간*을 지정 ('20~, 국토부) → '24년까지 전국 주요도로 구간에 완비

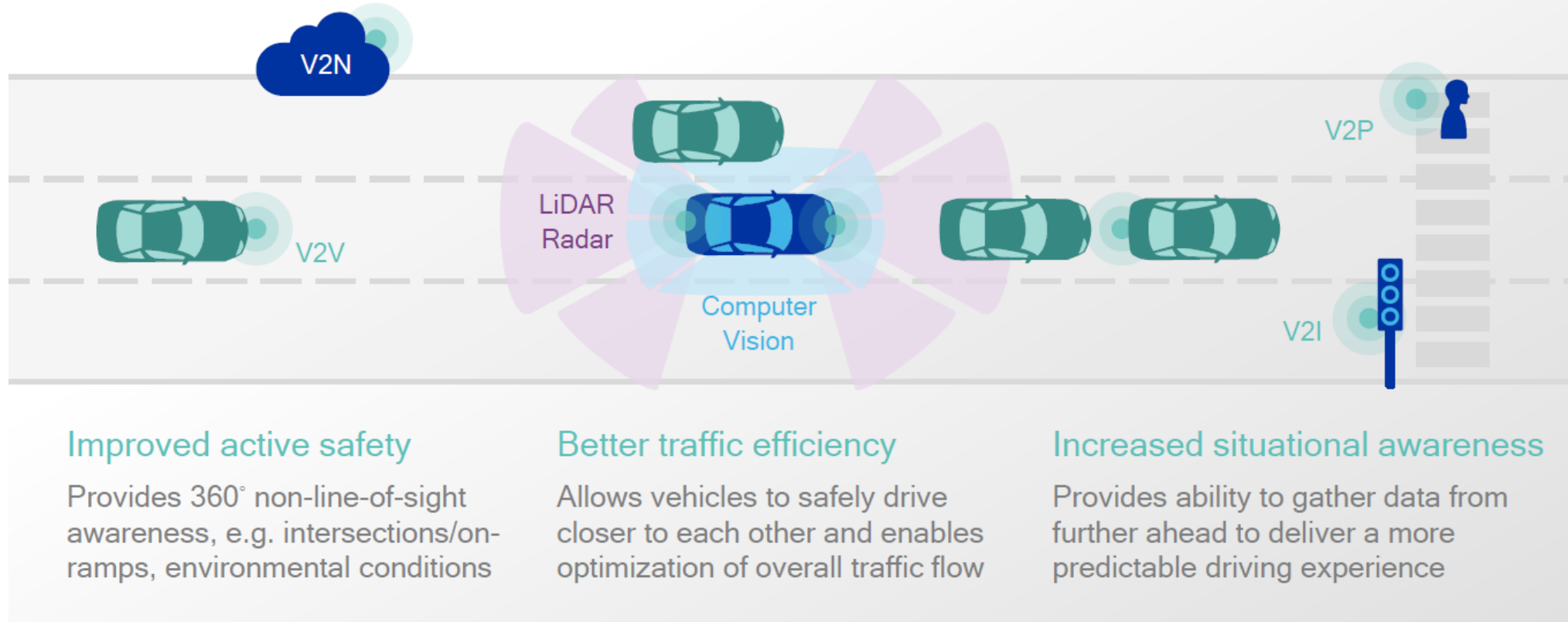
* 자동차전용도로 일정구간을 지정하여 우선 구축 (「자율차법」상 자율주행안전구간)

Our vision for the always-connected vehicle of the future
A safer, more efficient, more enjoyable driving experience



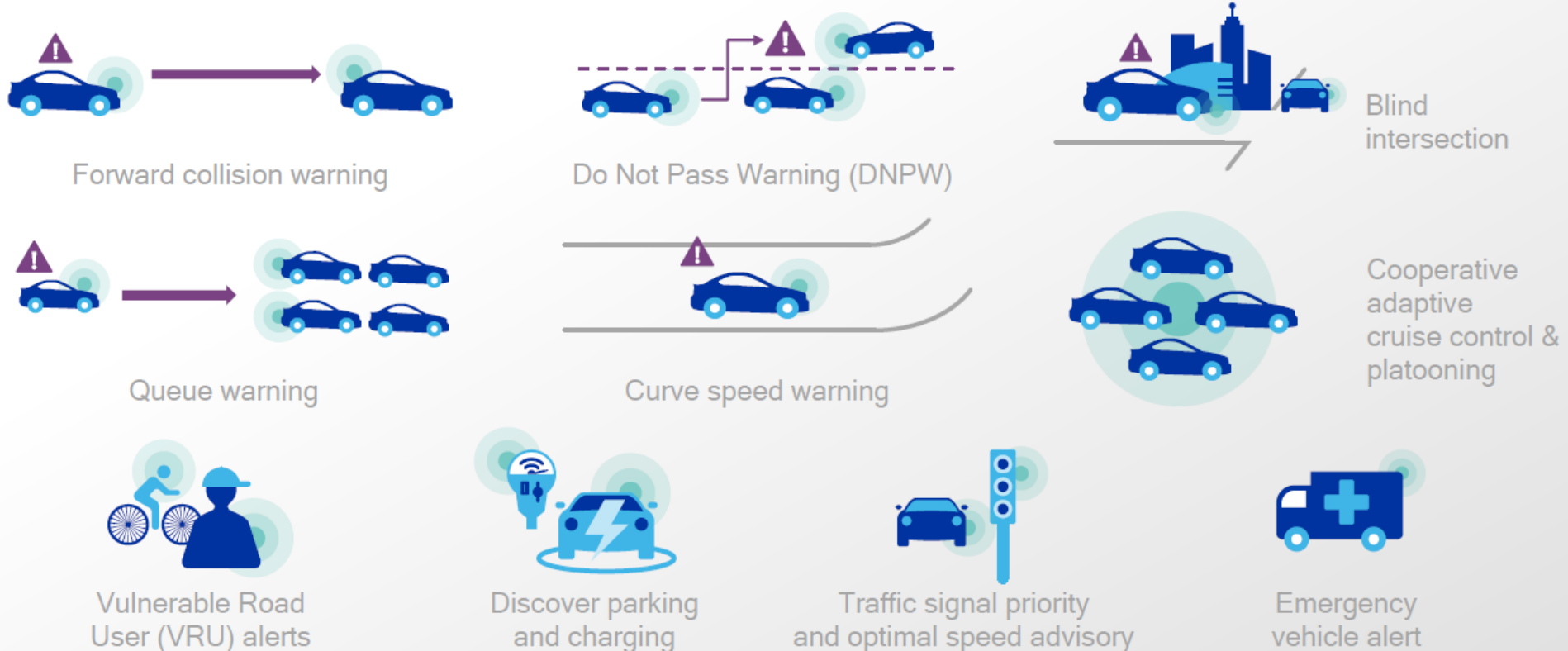
V2X is a key technology enabler to enhanced ADAS

Bringing significant value to Advanced Driver Assistance Systems (ADAS)

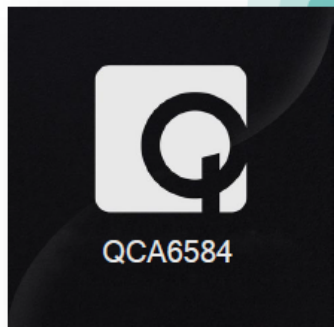


V2X enables a broad and growing set of use cases

Much more than collision avoidance



802.11p has established the foundation for V2X



2nd generation Qualcomm Technologies 802.11p offering with integrated Wi-Fi LAN and Bluetooth

Wi-Fi based technology - 802.11p standard

Adapted for latency-critical V2X communications in the 5.9 GHz band

Established security and upper layer specifications

With service layer / performance requirements defined by SDOs, e.g.SAE, ETSI-ITS¹

Path to DSRC² rulemaking in USA by NHSTA³ expected to start in 2016⁴

Based on 802.11p standard

Large scale field trials completed over the last decade

Commercially available technology here today

¹ Standard Development Organizations, e.g. Society for Automotive Engineers, European Telecommunications Standards Institute - Intelligent Transport Systems; ² Dedicated Short Range Communications (DSRC);

³ National Highway Traffic Safety Administration; ⁴ To improve road safety for future 'light vehicles' - Qualcomm has conducted extensive research into various use cases for DSRC, including V2P applications that could extend the safety benefits to vulnerable road users such as pedestrians and cyclists

Paving the path to more autonomous driving

Requires continued V2X technology evolution



Increasing safety requirements

Active safety use cases need to account for faster moving vehicles and denser traffic conditions



Expanding use cases

New situational awareness, traffic management, and connected cloud services



More vehicle data

From sharing simple status data today to a fully coordinated driving experience

Introducing Cellular V2X (C-V2X)

A unified connectivity platform for the connected vehicle of the future



Part of Release 14 of the global 3GPP standard

Target C-V2X specification completion end of 2016¹

Builds upon existing LTE connectivity platform for automotive

LTE already delivering key services today, e.g. telematics, eCall, connected infotainment

Enhances LTE Direct for V2X direct communications

Improvements over 802.11p - up to a few additional seconds of alert latency and 2x range²

Leverages existing LTE networks for V2X network communications

Using LTE Broadcast optimized for V2X to offer additional applications/services

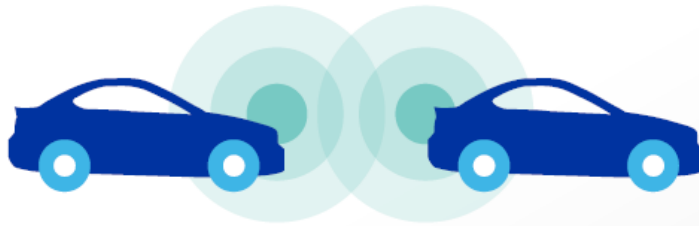
Rich roadmap towards 5G with strong ecosystem support

Technology evolution to address expanding capabilities/use cases

¹ For Direct communications component (enhancements to LTE Direct) - overall spec completion expected mid-2017; ² Based on Qualcomm Research simulations (see future slides for further information)

C-V2X defines two complementary transmission modes

PC5 interface
e.g. location, speed



Direct communications

Building upon LTE Direct device-to-device design with enhancements for high speeds / high Doppler, high density, improved synchronization and low latency

- Proximal direct communications (100s of meters)
- Operates both in- and out-of-coverage
- Latency-sensitive use cases, e.g. V2V safety

Uu interface
e.g. accident 1 kilometer ahead

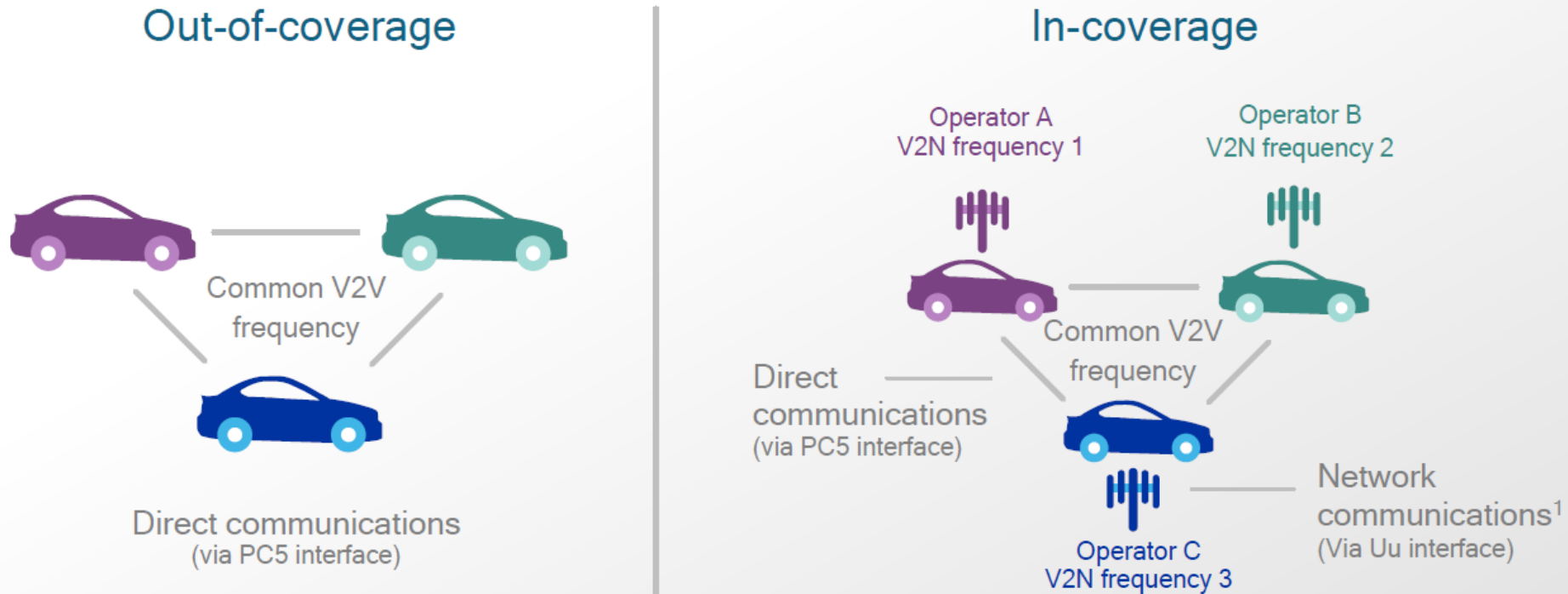


Network communications

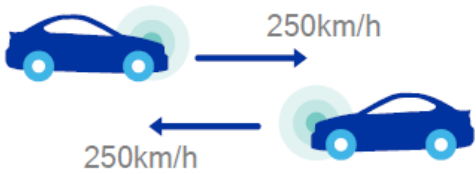
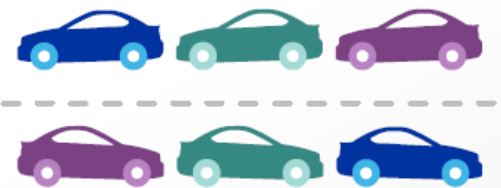
Using LTE Broadcast to broadcast messages from a V2X server to vehicles and beyond. Vehicles can send messages to server via unicast.

- Wide area networks communications
- Leverages existing LTE networks
- More latency tolerant use cases, e.g. V2N situational awareness

C-V2X designed for both in-coverage and out-of-coverage

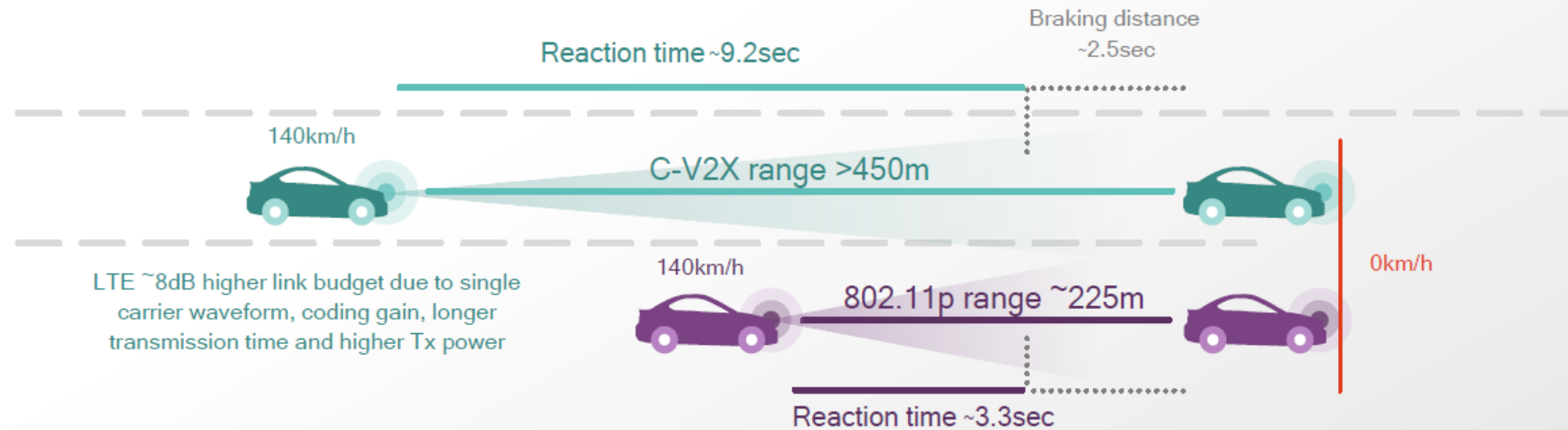


Overcoming the challenges of V2X communications

	V2X Challenges High relative speeds Leads to significant Doppler shift / frequency offset	C-V2X Solutions Enhanced signal design E.g. increasing # of ref signal symbols to improve synchronization and channel estimation
	V2X Challenges High node densities Random resource allocation results in excessive resource collisions	C-V2X Solutions Enhanced transmission structure Transmit control and data on the same sub-frame to reduce in-band emissions More efficient resource allocation New methods using sensing and semi-persistent resource selection
	V2X Challenges Time synchronization Lack of synchronization source when out-of-coverage	C-V2X Solutions Allow utilization of GPS timing Enhancements to use satellite (e.g. GNSS) when out-of-coverage

C-V2X increases reaction time over 802.11p/DSRC

For improved safety use cases - especially at high-speeds, e.g. highway



Safer driving experience

Increased driver reaction time

Support for high speeds

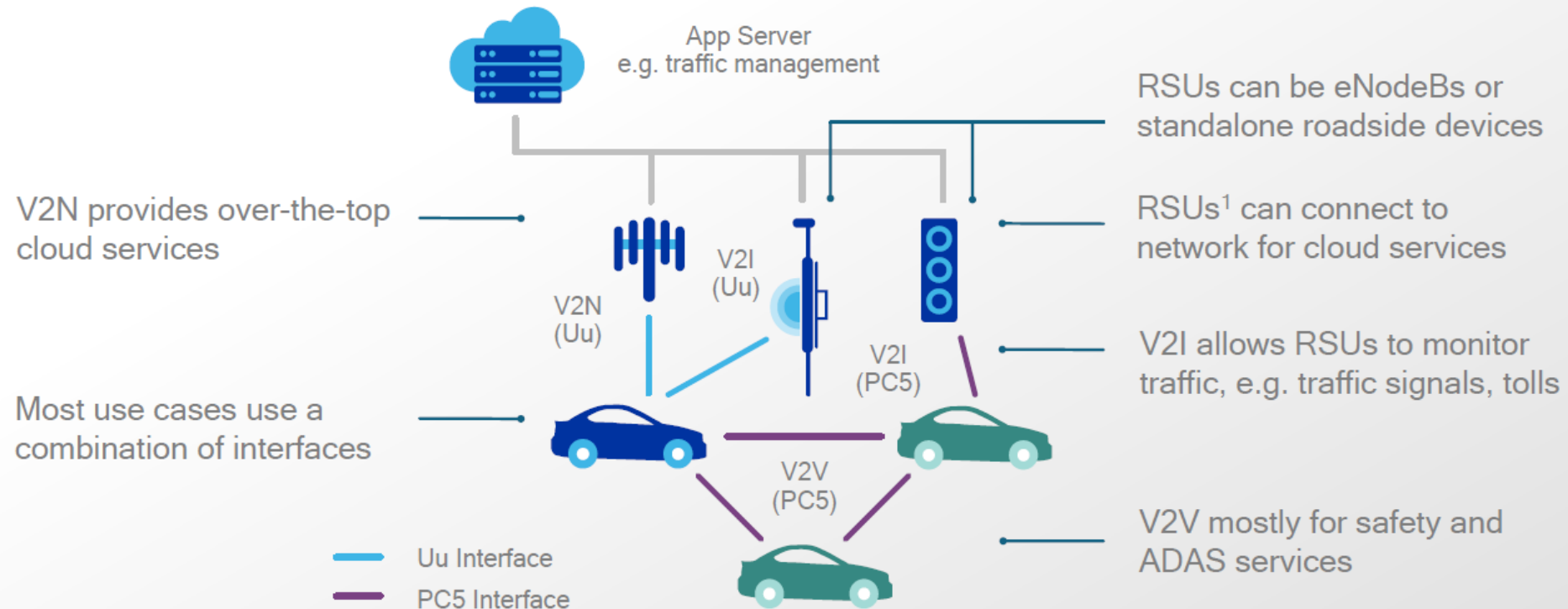
Relative speeds up to 500km/h

Increased situational awareness

Gather data from further ahead

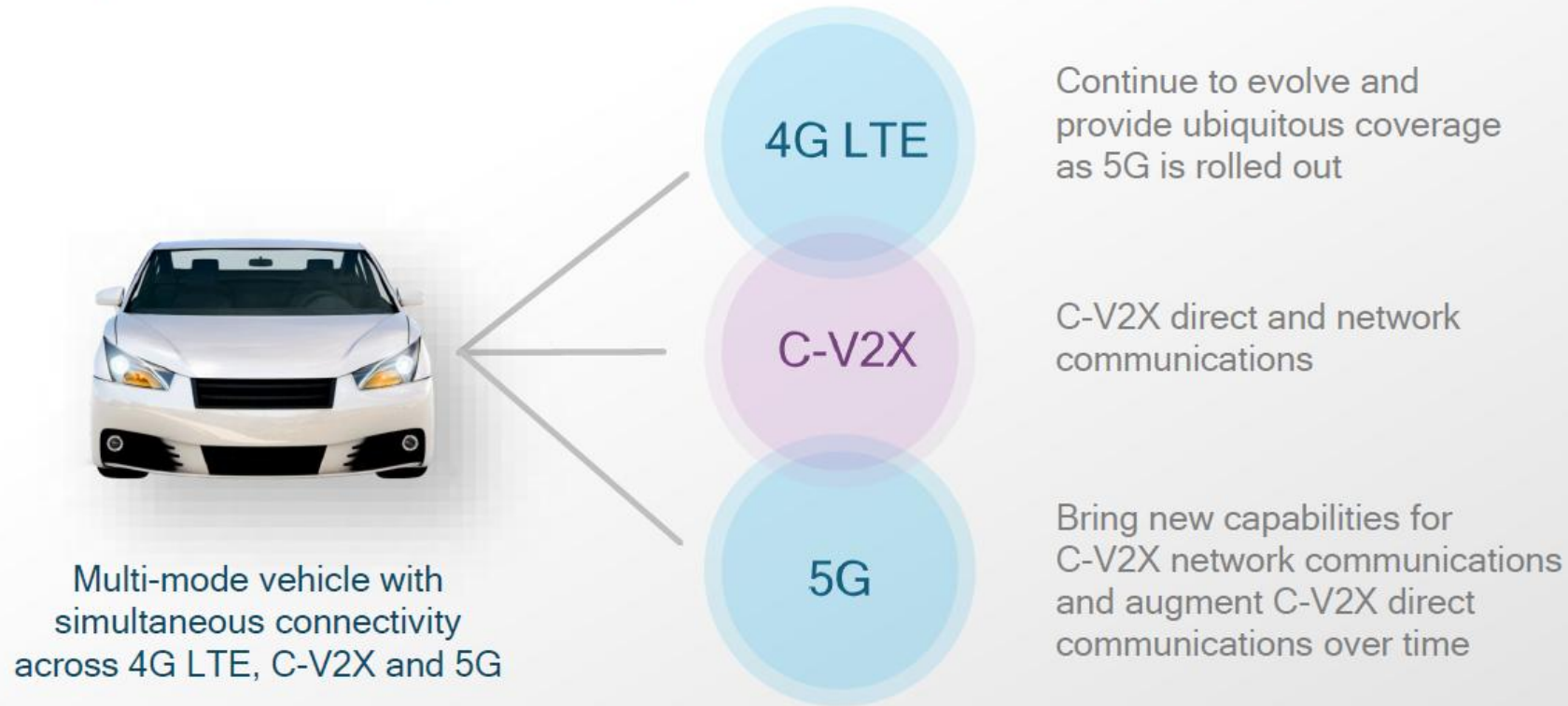
Delivering advanced services to vehicles

Opening up new opportunities and diverse business models for MNOs

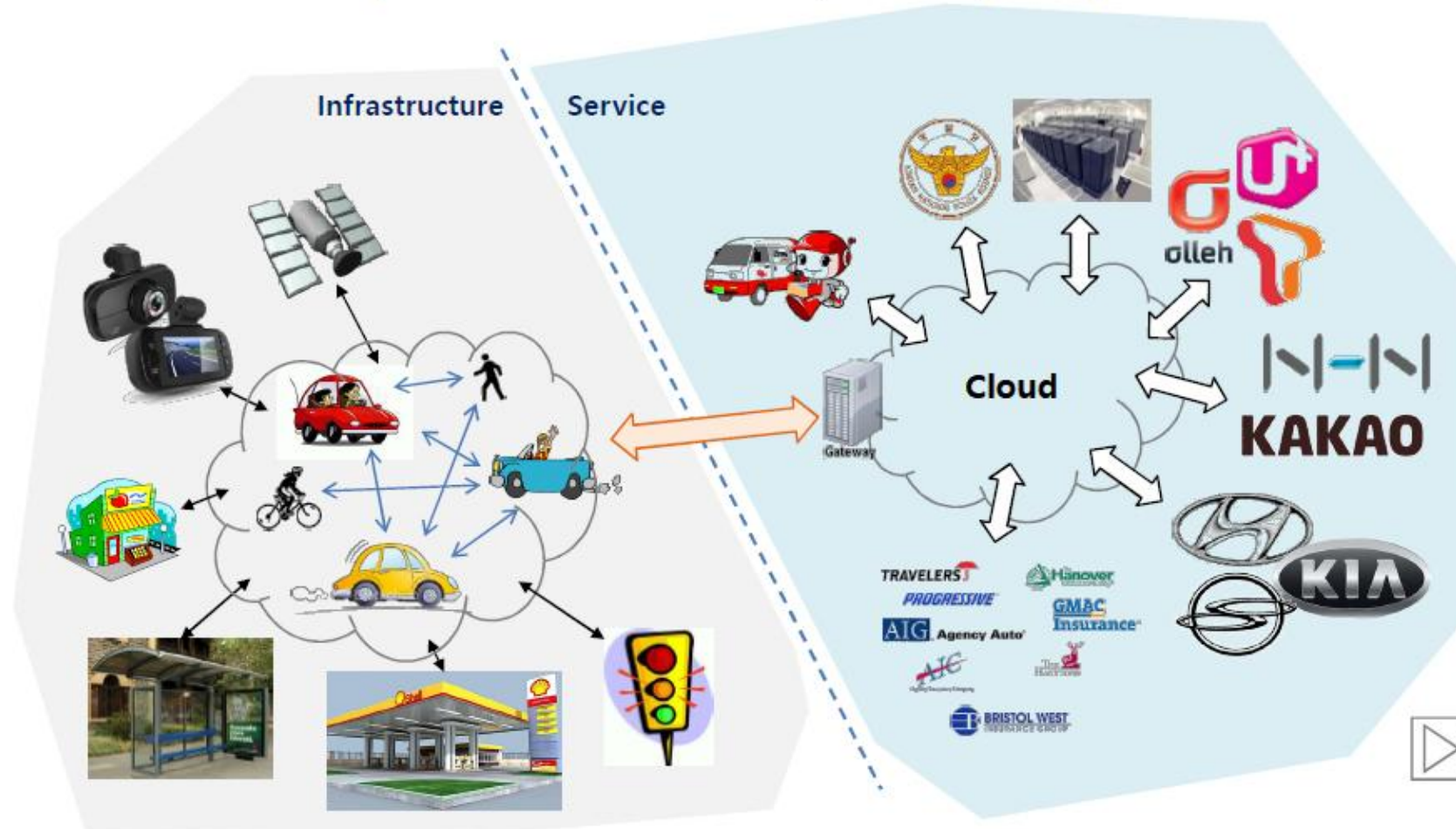


5G will build upon and enhance C-V2X

New 5G platform will augment / complement C-V2X—no 'rip and replace'

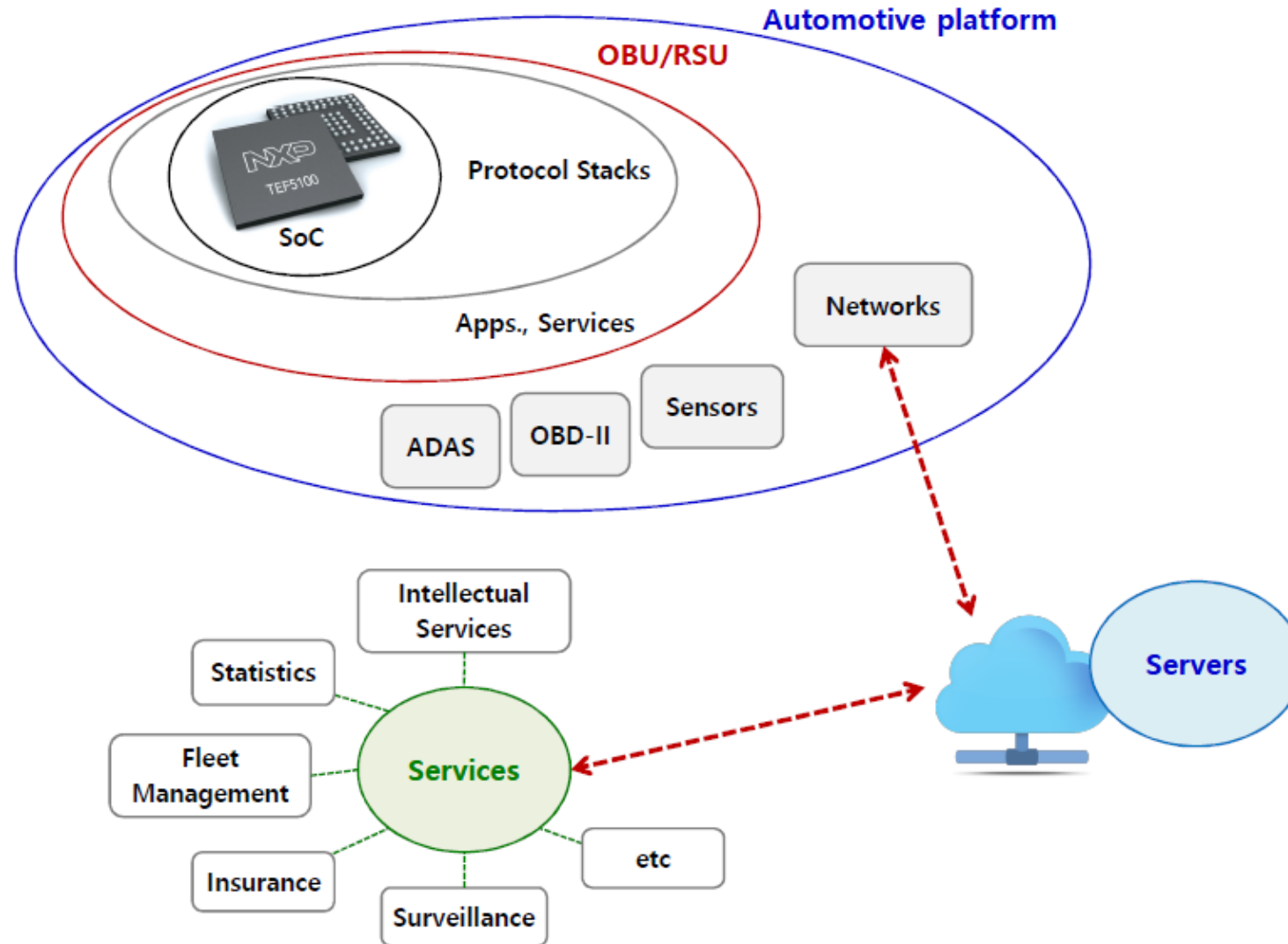


An **ecosystem** is a community of living organisms in conjunction with the non-living components of their environment, interacting as a system

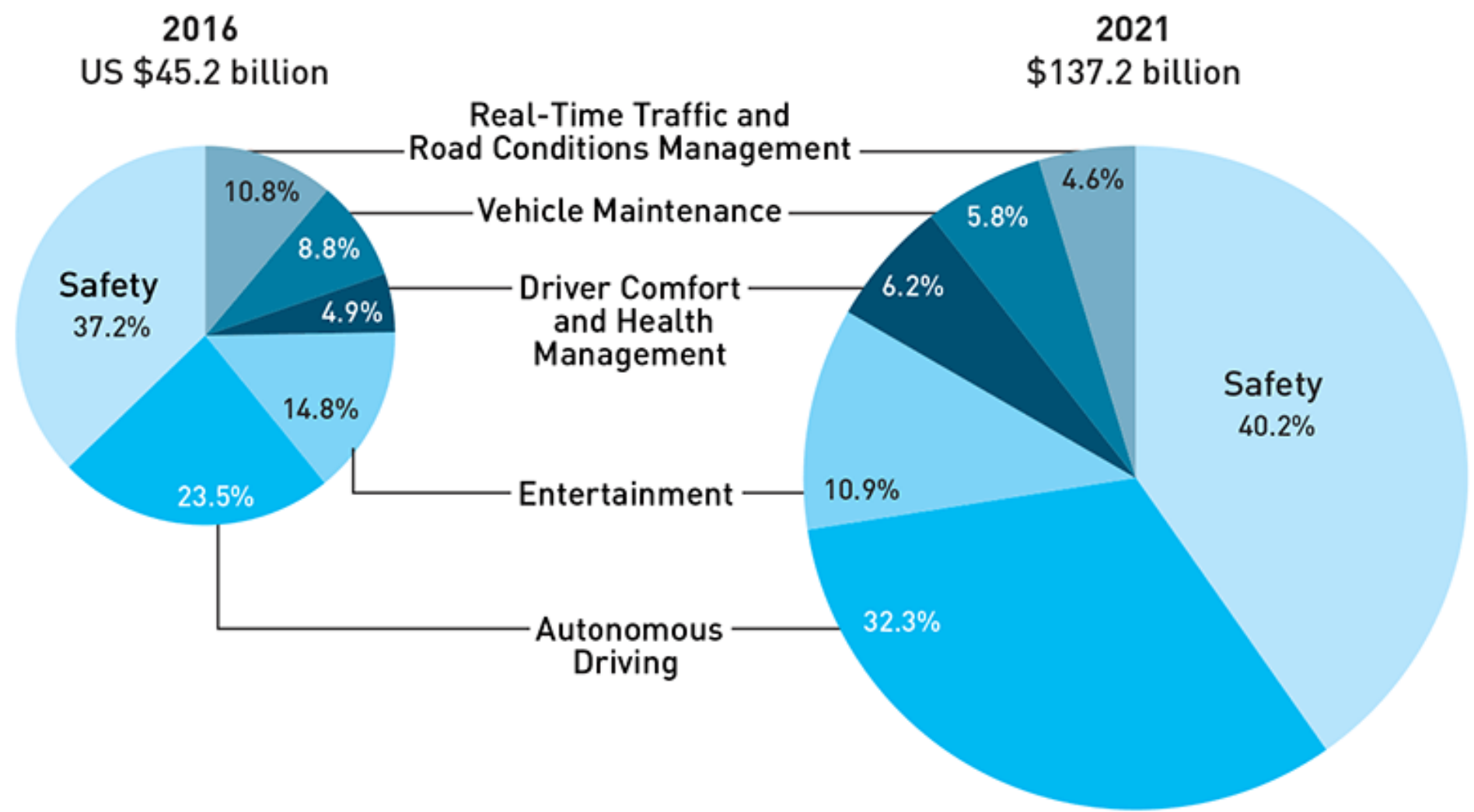


Connected Car Ecosystem

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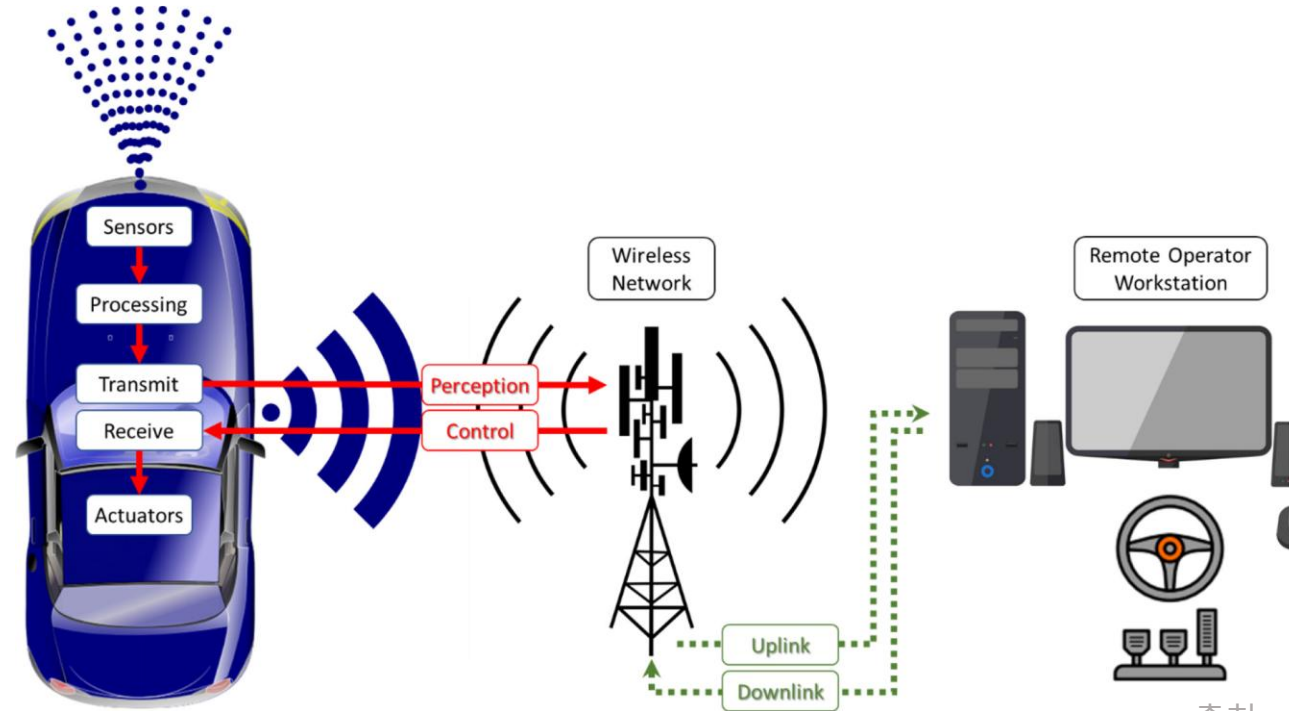
/ Connected Car Market



Why ToD ?

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- Wireless remote operation by a human is proposed as way to overcome the limitations of vehicle automation.
- When a Connected and Automated Vehicle (CAV) encounters a situation that the Automated Driving System (ADS) cannot safely manage, a remote operator could monitor, assist, or drive until the ADS can safely take control again.



출처 : BSI Standardizing remote operation vehicles

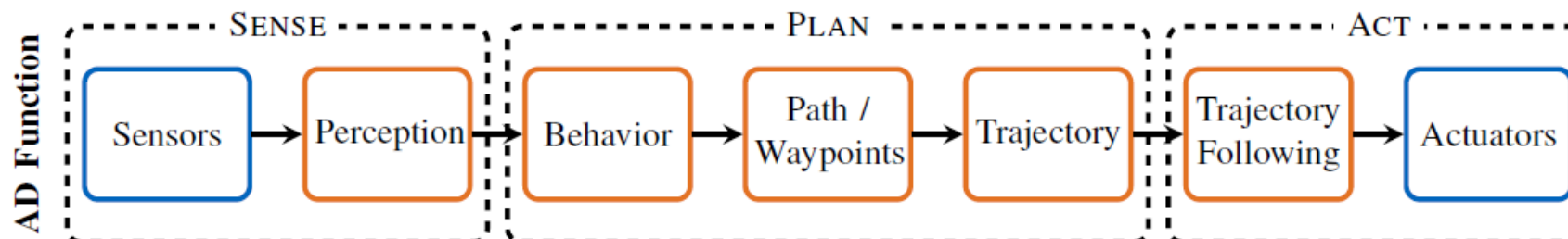
Use cases	Environment	Regulation	Speed	Carrying
Pavement robots	Uncontrolled	Off-road	Very low	Goods only
Local delivery robots	Uncontrolled	On-road	Medium	Goods only
Buildings & factories	Controlled	Off-road	Low	Goods only
Yards & warehouses	Controlled	Off-road	Low-Medium	Goods & machinery
Construction & agriculture	Controlled	Off-road	Low	Machinery
Rental car deliveries	Uncontrolled	Off-road	Road speeds	Empty
Remote valet parking	Un/Controlled	Off-On-road	Low-Medium	Empty
CAV safety backup	Uncontrolled	On-road	Road speeds	Passenger 
CAV truck safety backup	Uncontrolled	On-road	Road speeds	Goods

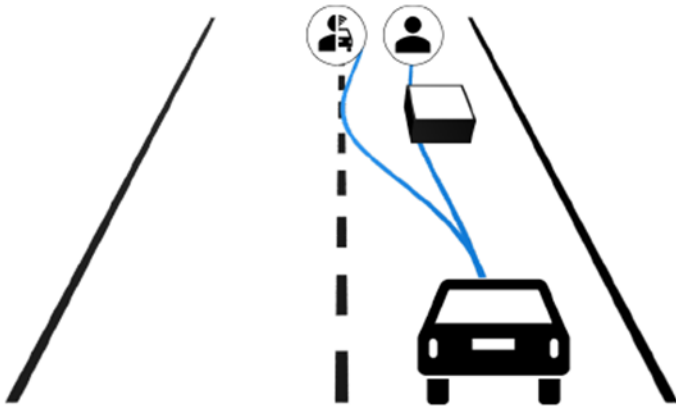
출처 : BSI Standardizing remote operation vehicles

/ Function Description

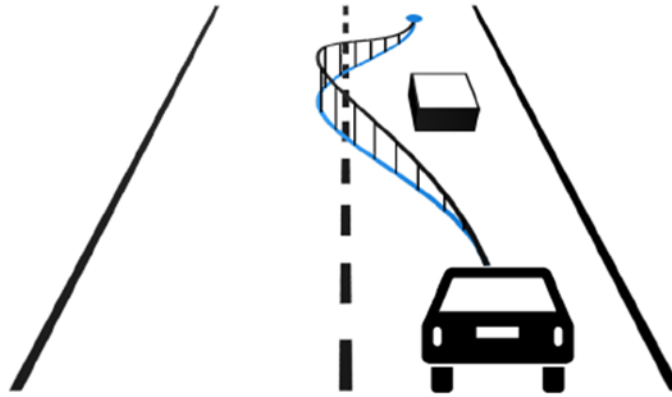
Term	Description	Function	Input Devices
Remote Driving	System is fully under remote control.	Direct Control	S & P
		Shared Control	S & P
		Trajectory Guidance	S & P, M & K
Remote Assistance	System receives event-driven remote assistance from the operator, while still being responsible for the driving task.	Waypoint Guidance	M & K
		Interactive Path Planning	M & K
		Perception Modification	M & K
Remote Monitoring	System is remotely monitored with very limited intervention possibilities.	-	-

S & P : Steering and Pedals
M & K : Mouse and Keyboard

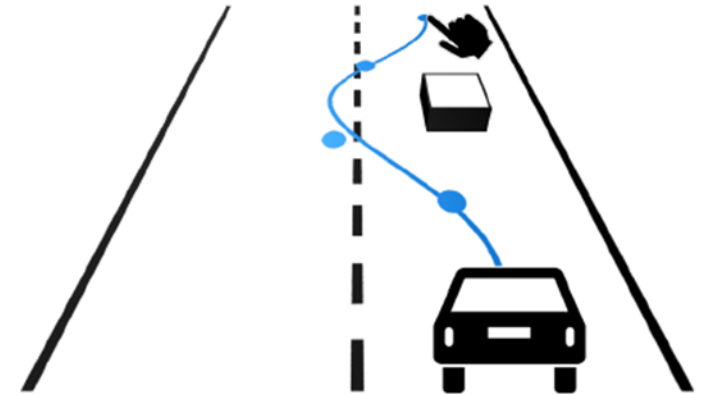




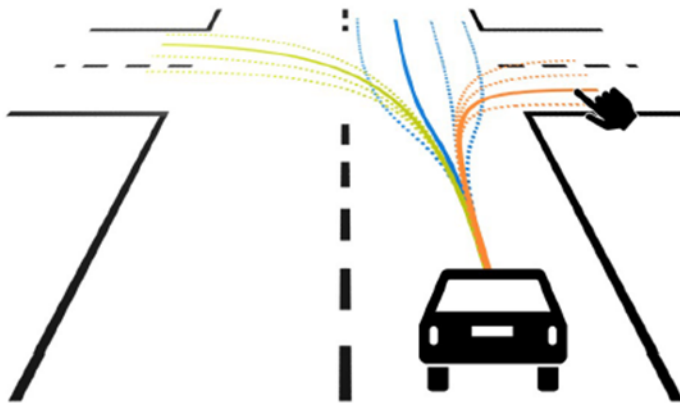
Shared Control



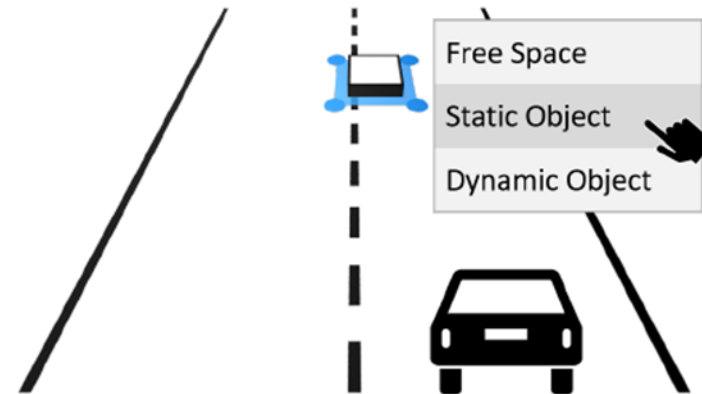
Trajectory Guidance



Waypoint Guidance

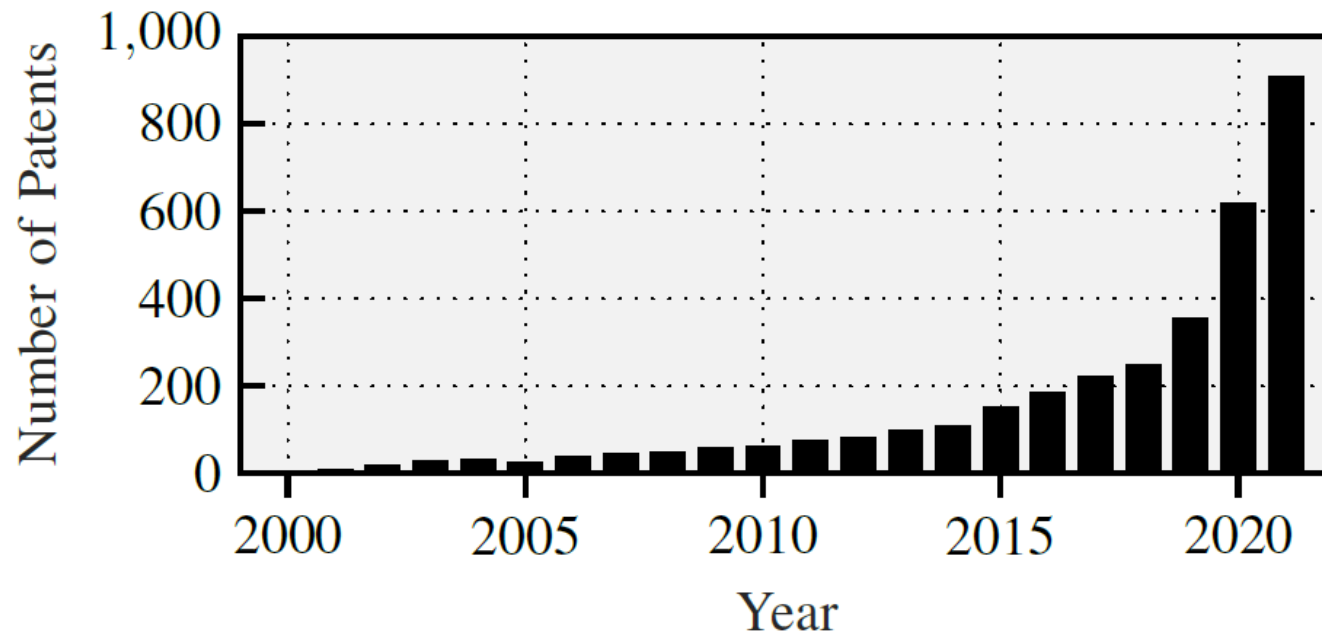


Interactive Path Planning



Perception Modification

- For a long time, remote control of AVs was reserved for specific operational scenarios, usually in locations that would be dangerous and hazardous for manual operation by humans.
- However, finally it became evident that no matter how good full autonomy is, self-driving vehicles will always require some kind of remote monitoring, assistance, or driving to overcome challenging traffic scenarios.



출처 : Google patent search

- Remote technology for **logistics vehicles** that enhances safety, productivity, and operational efficiency across the supply chain
- One Platform, Multiple Logistics Solutions



Virtualize your forklift operations

Enable a workforce of "digital drivers" to remotely operate vehicles from up to thousands of miles away. Improve safety, expand access to labor, and unlock productivity with fully remotely-operated forklifts.

Remotely Operated Forklifts

Scale autonomy faster

Remotely resolve edge cases and expand operational capabilities for autonomous vehicles. Unlock maximum uptime and continuous system improvements by combining automated productivity with human adaptability.

Assisting Autonomy



Onboard and train drivers from anywhere

Remotely onboard and train new drivers from anywhere with real-time telepresence to reduce safety risks and improve driver retention.

Distanced Driver Training

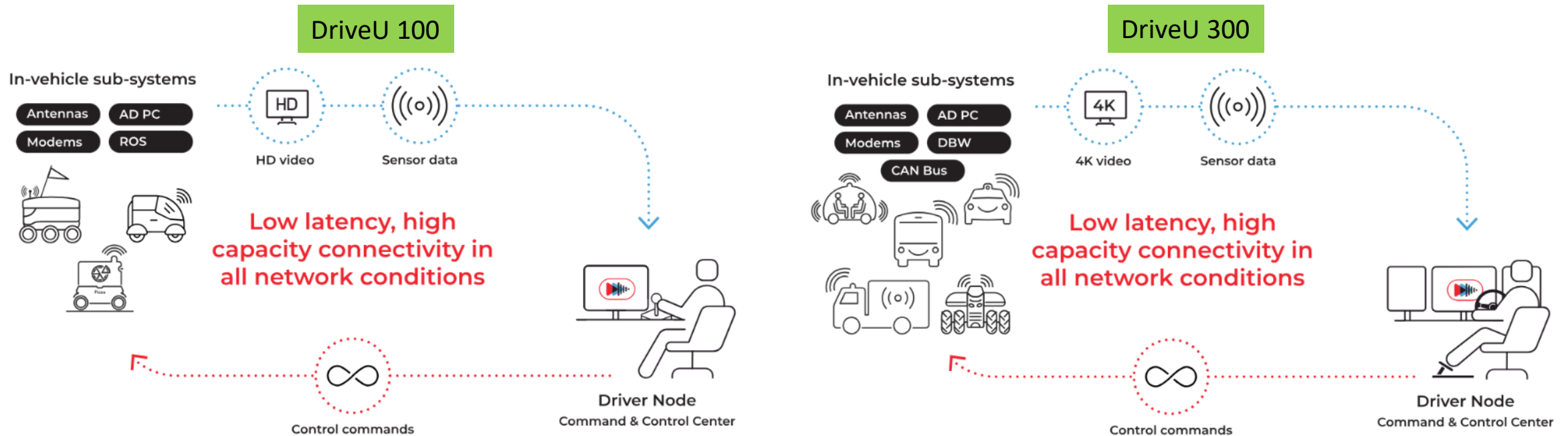
/ Ottopia (<https://ottopia.tech/>)

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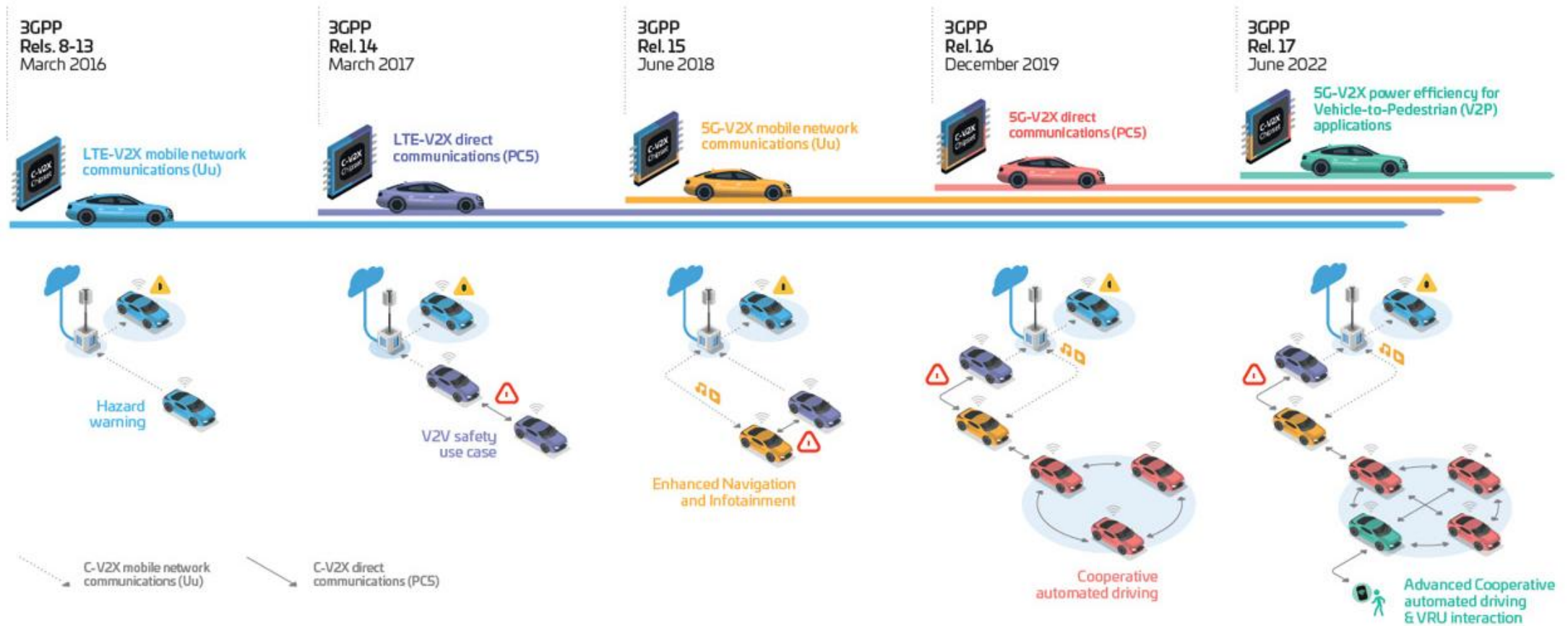
- Ottopia offers the best teleoperations **software platform** on the market



- Deploy **robot** and **autonomous vehicle** fleets with confidence, using teleoperation
- Superior teleoperation connectivity platform for **low latency video**

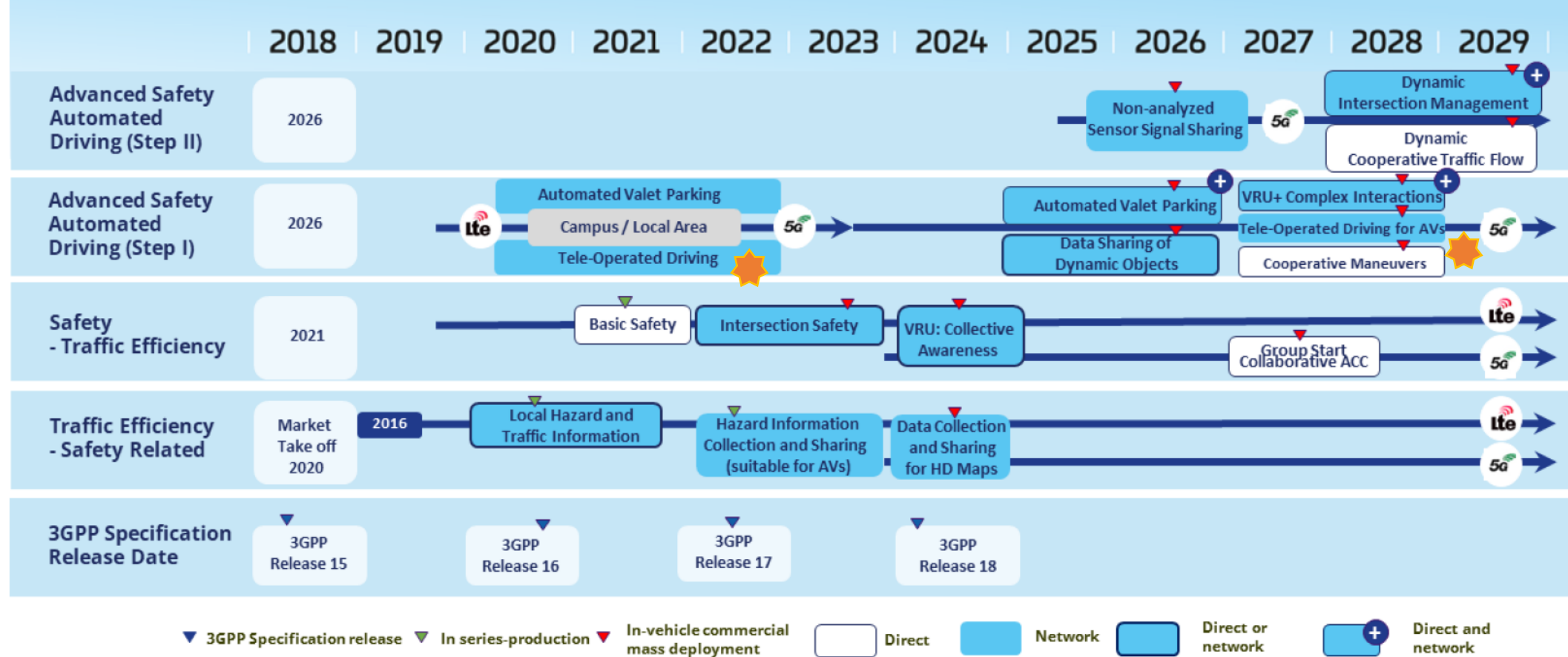


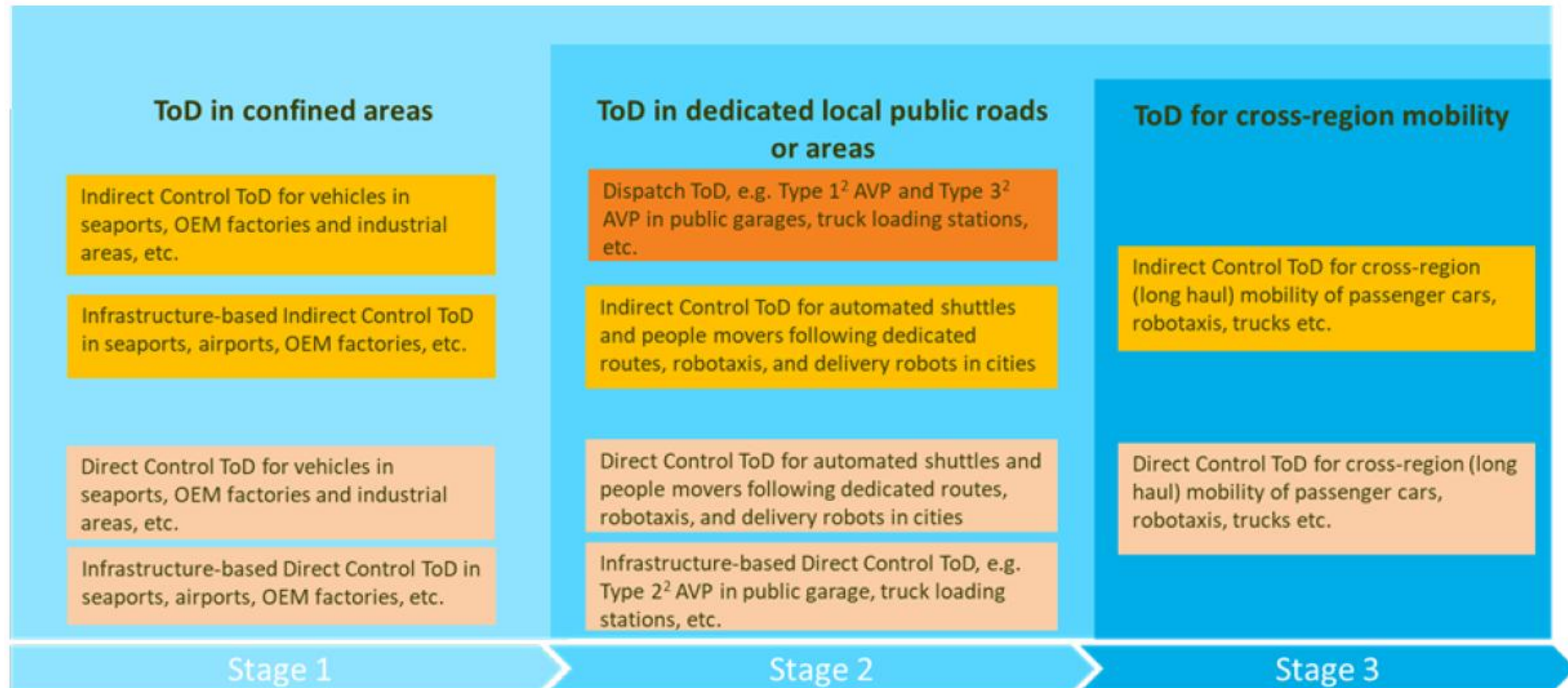
5GAA - 3GPP Roadmap



5GAA(5G Automotive Association), 3GPP(3rd Generation Partnership Project)

Expected timelines for mass deployment of C-V2X use cases





/ Network Considerations

Information Type	Data Rate	Note
Uplink (vehicle to ToD operator)		
Video (for human ToD operator)	Up to 32 Mb/s	From the vehicle to the ToD operator: ~8 Mbps are needed for a progressive high-definition video/camera (h.264 compression), assuming four cameras are needed (one for each side). [2] Reliability: 99%
Object information	Up to 4 Mb/s	Sensor data (interpreted objects) can also be provided from the vehicle to the ToD operator, assuming 1 kB/object/100 ms and 50 objects. [2] Reliability: 99%
Audio	~96 kb/s	Audio signal provides acoustic environmental information to the ToD operator. Audio signal may be embedded in the video stream. This signal can also be used for the ToD operator to have interactive audio communication with the passengers onboard.
Vehicle information (e.g. speed, acceleration, vehicle position)	~0.2 Mb/s	50 km/h is considered the maximum speed for remote steering under highly uncertain conditions. In the ToD support case or in ToD for automated parking use cases, the vehicle speed is assumed to be no more than 10km/h and 20 km/h, respectively.

Downlink (ToD operator to vehicle)		
Vehicle manoeuverer commands (Direct Control ToD type)	~ 400 kb/s	The size of command messages, e.g. a) turn steering wheel, direction, angle, b) apply the brake, brake pressure, etc. including appropriate security headers. The command messages will be sent every 20 ms (maximum 50 messages per second) [2] Reliability: 99.9% or higher
Driving path or trajectory (Indirect Control ToD type)	~ 25 kb/s	Data from the provided paths are several kbps (e.g. 100 points and 32 bytes for each point). [2] Reliability: 99.9% or higher
Keep alive signal	~0.2 Mb/s	An example of this signal is the cyclic status message from Automated Valet Parking System (AVPS) to the vehicle for the AVP use case. [9] This signal is only needed when there is no periodic high frequency transmission of driving instruction commands from the ToD operator to vehicle, e.g. in Indirect Control ToD.
Control signals (e.g. for camera setting)	negligible	This signal is not a continuous data stream and is considered as a very low data rate transmission.
Video/Audio (communication with ToD operator)	Video ~ 8 Mb/s Audio ~ 96 kb/s	These signals are applicable only when the human ToD operator needs interactive audio and/or video communication with the passengers in the vehicle.

/ Latency Requirement from 5GAA

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- For the Tele-operated Driving use case (vehicle speed up to 50 km/h)

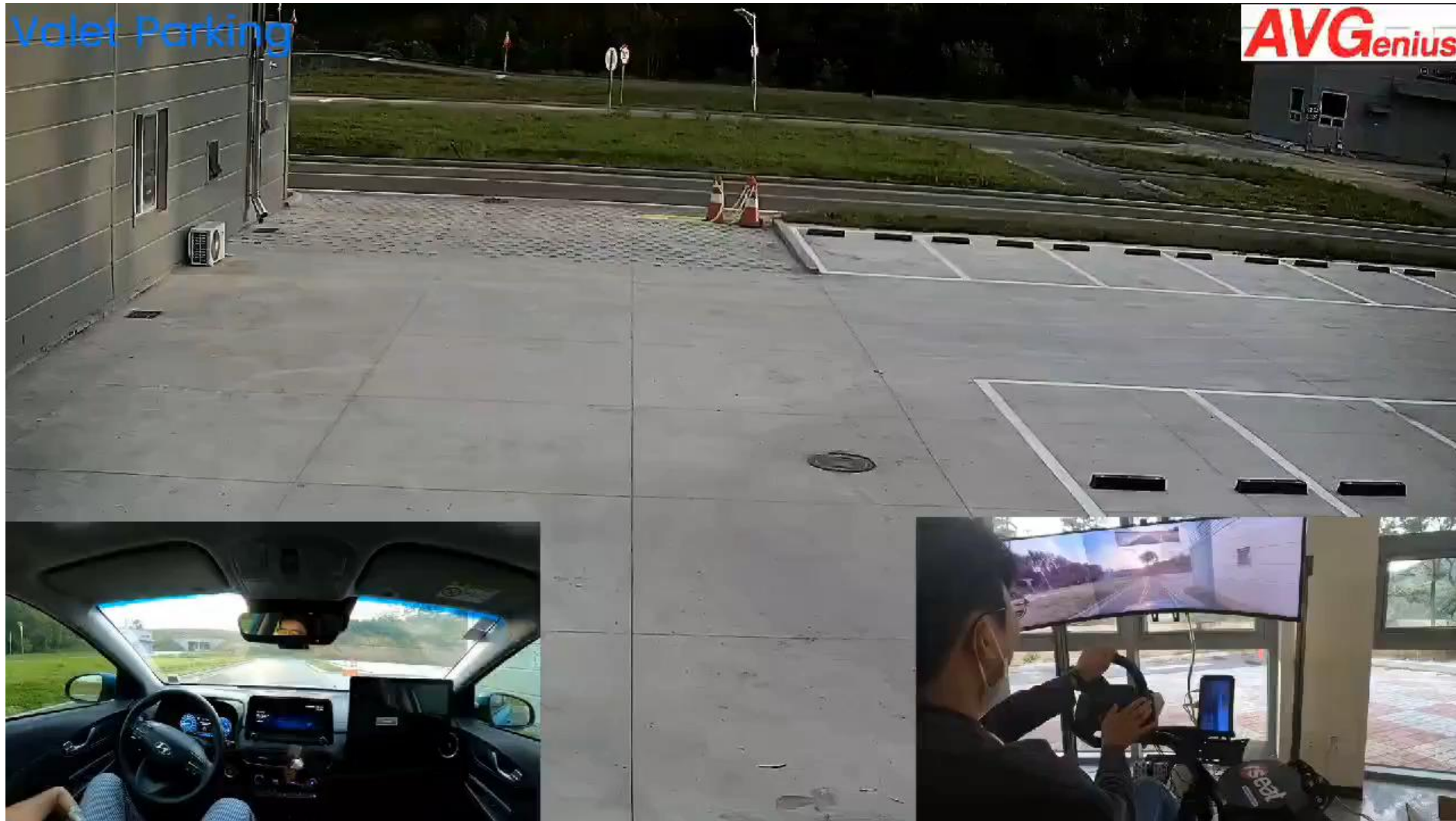
Service level latency	[ms]	• From HV to ToD operator: 100 • From ToD operator to HV: 20	From ToD operator to HV: Depends on the reaction time needed, which is directly related to the maximum driving speed allowed. For instance, at a speed of 50 km/h, the HV will move 0.27 m within 20 ms.
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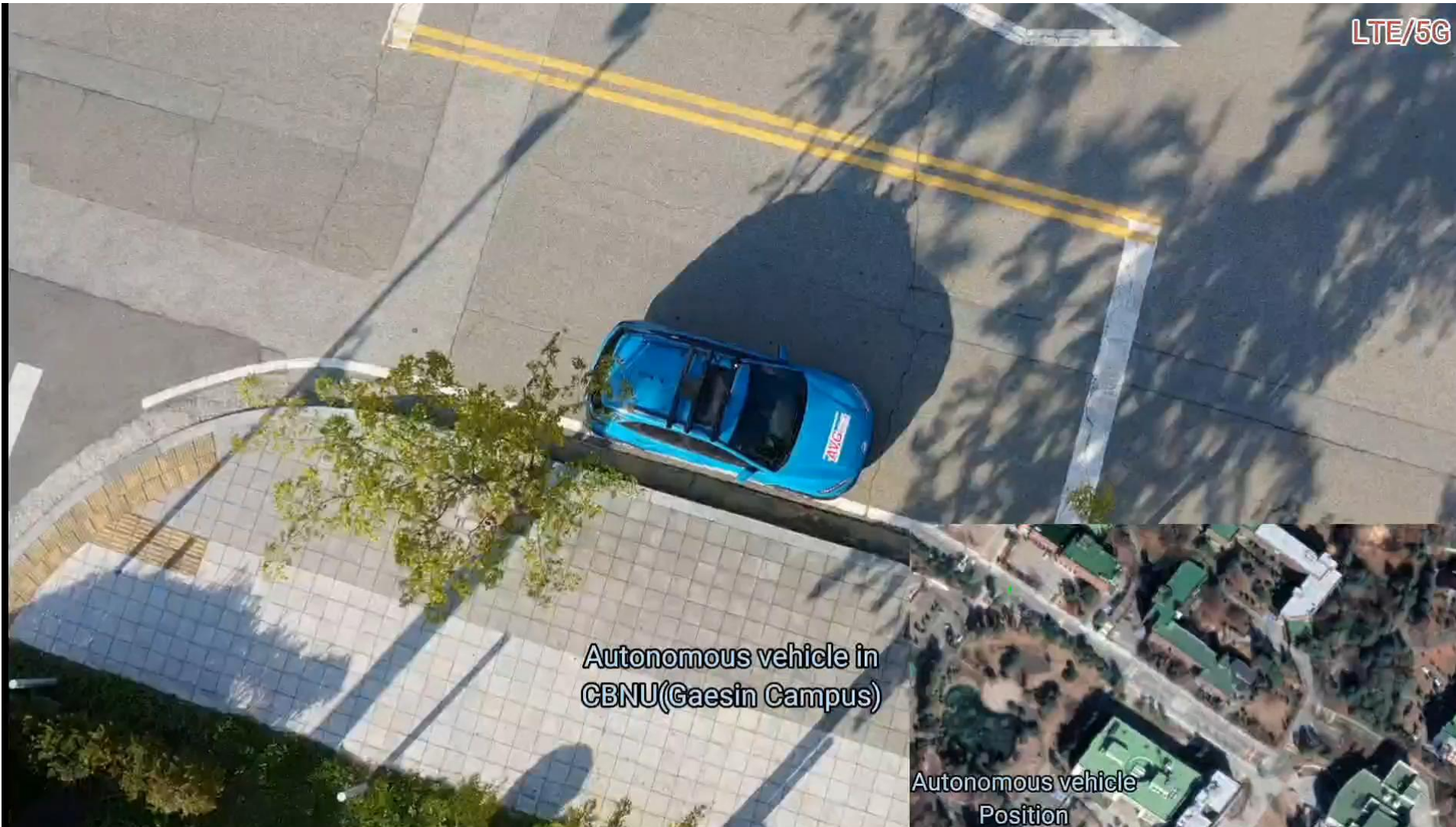
- For user story #1 of Tele-Operated Driving Support: Remote Steering (Direct Control ToD, vehicle speed up to 10 km/h)

Service level latency	[ms]	• From HV to ToD operator: 100 • From ToD operator to HV: 20	From ToD operator to HV: Depends on the reaction time needed, which is directly related to the maximum driving speed allowed.
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- For user story #2 of Tele-Operated Driving Support: Remote Driving Instructions (Indirect Control ToD, vehicle speed up to 10 km/h)

Service level latency	[ms]	• From HV to ToD operator: 100 • From ToD operator to HV: 200	From ToD operator to HV: With only the instructions to be transmitted from ToD operator to the HV, latency requirements are more relaxed.
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Thank you for Attention

Q & A

